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Machine Learning for Postoperative Outcomes Prediction

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Abstract

OBJECTIVE: Accurate prognosis of postoperative recovery is essential for hospital resource management and personalized rehabilitation. This study aimed to develop and validate machine learning (ML) models to predict the length of stay (LoS), social discharge destination, and functional recovery (Barthel Index) in patients undergoing hip or knee fracture surgery.

METHODS: A retrospective cohort of 1,710 patients (year 2019) from the Galeazzi Orthopedic Institute (Milan, Italy) was analyzed. The feature space was expanded to 77 predictors, including demographic, clinical, surgical, and early functional data (BI/FIM). Preprocessing included Multiple Imputation by Chained Equations (MICE) to handle missing pharmaceutical data (21.8%). Several architectures were evaluated: Ridge/Lasso, Random Forest (RF), LightGBM, Support Vector Machines (SVM), and Multi-Layer Perceptron (MLP). Internal validation was performed via 5-fold cross-validation, while a temporal external cohort (year 2018, n=360) was used to assess generalizability. Explainability was ensured through SHAP values and permutation importance.

RESULTS: In internal validation, non-linear models significantly outperformed linear baselines ($p_{\text{corr}} < 0.05$, Wilcoxon test). For the primary outcome (LoS), SVR and LightGBM achieved the highest accuracy ($R^2 = 0.265$ and 0.250; MAE = 2.20 days). Social outcome prediction reached a Balanced Accuracy of 0.856 and an F_2-score of 0.656 using LightGBM with cost-sensitive learning. Key predictors across all tasks included basal functional scores (BI/FIM), surgery duration, age, and preoperative hemoglobin. However, external testing revealed a severe performance collapse ($R^2 = -0.027$ for LoS). Adversarial validation (Accuracy = 1.0) and Kolmogorov-Smirnov tests ($p = 0.0$ for surgery duration and hospital LoS)

confirmed a significant temporal data shift. A “cohort mixing” strategy (n=1,963) restored clinical utility, yielding an R^2 of 0.253 and a ROC-AUC of 0.84 for social outcome.

CONCLUSIONS: While ML models demonstrate a robust ability to capture complex rehabilitation patterns internally, temporal data shift remains a critical barrier to deployment. The integration of multi-year datasets through cohort mixing proved to be the only effective strategy to mitigate concept and label drift. These results support the implementation of semi-annual model maintenance protocols to ensure stable Clinical Decision Support Systems in orthopedic settings.

Keywords: Machine Learning, Orthopedic Rehabilitation, Length of Stay, Data Shift, Explainable AI, Clinical Decision Support System.

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Chapter 1

Introduction

1.1 Context and Motivation

The progressive aging of the global population has led to a substantial increase in orthopedic and trauma-related conditions, particularly hip and knee fractures among elderly individuals. The *Istituto Ortopedico Galeazzi* in Milan, an institution of high clinical specialization, provided the data foundation for this study, which analyzes a cohort of 1,710 patients who underwent surgery during the year 2019. This dataset reflects a clinically complex population with a mean age of 69.4 years (± 10.73 SD) and a median age of 70 years (IQR: 15). Gender distribution shows a predominance of female patients (62.34%), consistent with the higher prevalence of bone fragility in this subgroup. Furthermore, the cohort is characterized by chronic comorbidities, including arterial hypertension (57.8%), thyroid diseases (13.9%), type 2 diabetes mellitus (11.2%), and cardiovascular conditions (11.6%). From a socio-residential perspective, 59.3% of individuals are retired and 81.3% live with family members, a factor identified as relevant for successful protected discharge outcomes.

The primary motivation of this work is to optimize rehabilitation planning and hospital resource management. Predictive modeling is employed to support clinical decision-making across three domains. First, estimating the rehabilitation length of stay enables more efficient bed management. Second, functional outcomes are assessed through the prediction of post-operative autonomy using the Barthel Index (BI) and the Functional Independence Measure (FIM). Third, early identification of patients at risk of losing residential independence provides insights for discharge planning. The adoption of advanced Machine Learning (ML) and

Deep Learning techniques is justified by the non-linear nature of clinical features. Preliminary analyses indicate that simple linear associations are generally weak; for instance, the correlation between surgical duration and length of stay is $r = 0.35$, while family support shows a negative correlation with the risk of non-protected discharge ($r = -0.29$). Consequently, ensemble methods and kernel-based approaches are required to capture complex interactions that traditional linear models fail to represent effectively.

1.2 Thesis Objectives

The present thesis aims to develop, validate, and optimize predictive models for post-operative outcomes using the previously described cohort. Beyond accuracy, the study is designed to deliver a Clinical Decision Support System (CDSS) that is robust to temporal variations and interpretable for clinical personnel. In alignment with management priorities, the primary outcome is formulated as a regression task for predicting rehabilitation length of stay, which exhibits a mean of 10.28 days (SD: 3.72). Secondary outcomes include a binary classification for social outcomes, distinguishing between autonomous home discharge and discharge requiring institutional or family care, and the estimation of care burden, where 76.4% of patients require fewer than three hours of daily assistance. Functional recovery is also addressed through regression tasks predicting post-operative BI score.

From a methodological perspective, the thesis addresses several computational challenges inherent to real-world clinical data. A core objective is the mitigation of data shift, which was identified when model performance collapsed on an external test set from the year 2018. To address this, the study investigates multiple strategies, including calibration techniques such as Platt Scaling and Isotonic Regression to reduce the Expected Calibration Error (ECE). Furthermore, the work aims to resolve issues related to missing data and class imbalance. This involves implementing iterative imputation via MICE for variables with high missingness, such as pharmacological therapy (21.81%), and applying balancing techniques like SMOTENC and weighted loss functions. Finally, the study seeks to ensure statistical rigor through non-parametric hypothesis testing with Benjamini–Hochberg corrections and to integrate a Green AI perspective by monitoring the carbon footprint of the training process.

1.3 Contribution of the Work

The primary contribution of this thesis lies in the development of an end-to-end analytical framework that overcomes the limitations of traditional clinical statistics. From a methodological standpoint, the work introduces a structured pipeline tailored to clinical complexity. The implementation of Multiple Imputation by Chained Equations (MICE) is empirically validated through systematic comparisons; for instance, the Support Vector Regression (SVR) model achieved an R^2 of 0.265 with imputation, compared to 0.216 without. Similarly, the use of SMOTENC and BalancedRandomForest successfully addressed the heavy class imbalance in social outcome variables, where 90.6% of observations belonged to the majority class.

A key scientific contribution concerns the identification and mitigation of temporal data shift. Detection was performed using adversarial validation, which achieved an accuracy of 1.0 in distinguishing between the 2018 and 2019 datasets, confirming a substantial distributional discrepancy. To resolve this, cohort mixing was explored, successfully transforming a non-informative predictive performance ($R^2 = -0.027$) into a robust model with an R^2 of 0.253 for the primary outcome. Quantitative results indicate that ensemble methods like LightGBM and Random Forest significantly outperform regularized linear models like Ridge and Lasso. These differences are statistically validated using Wilcoxon and Mann–Whitney tests ($p_{\text{corr}} < 0.05$). To ensure clinical applicability, the framework incorporates SHAP values to provide transparent insights into feature impact, such as the influence of hemoglobin levels on length of stay. In conclusion, this thesis establishes a comprehensive protocol for addressing clinical variability, advancing the system toward a Technology Readiness Level (TRL) of 4 [1].

Chapter 2

Literature Review

2.1 Machine Learning in Clinical Data Science

The integration of Machine Learning (ML) into clinical practice represents a substantial paradigm shift from traditional statistical approaches toward predictive modeling frameworks capable of handling the high dimensionality of modern healthcare data. Contemporary clinical data are inherently complex, encompassing structured Electronic Health Records (EHRs), unstructured clinical narratives, medical imaging, and, increasingly, high-throughput biological data. This heterogeneity has challenged conventional modeling techniques and motivated the adoption of more flexible and scalable ML methodologies [2].

Historically, clinical decision-making has relied on simplified scoring systems, such as the Apgar score or the ASA physical status classification [3]. While these approaches provide interpretable and robust assessments, they are inherently limited in capturing complex, non-linear relationships among physiological variables. In contrast, ML models enable the extraction of latent patterns from longitudinal patient histories and high-dimensional feature spaces, supporting more accurate and personalized clinical decision-making.

A key task in leveraging EHR data is clinical phenotyping, defined as the identification of patients with specific conditions based on their digital records. Traditional phenotyping approaches relied on expert-defined rule-based systems, which, although precise, were time-consuming to develop and difficult to generalize across institutions. More recent approaches employ supervised and weakly supervised ML techniques, leveraging proxy labels derived from diagnostic codes or natural language processing outputs to scale phenotype identification

[4].

In contemporary clinical applications, ML methodologies are primarily employed for diagnostic support, outcome prediction, and risk stratification. Within orthopedic and musculoskeletal domains, convolutional neural networks (CNNs) have demonstrated strong performance in the automated analysis of radiographic images, including fracture detection and disease severity grading [5, 6]. At the same time, predictive models based on structured clinical and socioeconomic variables are increasingly used to estimate post-operative outcomes, such as readmission risk [7] or prolonged hospitalization, thereby supporting personalized perioperative management strategies.

The selection of ML architectures in clinical research is typically governed by the trade-off between predictive performance and interpretability. Deep learning models are particularly effective in image-based and unstructured data settings, where they can automatically learn hierarchical feature representations. In contrast, structured tabular data derived from EHRs are more commonly modeled using ensemble techniques such as Random Forests [8] and gradient boosting methods, including Extreme Gradient Boosting (XGBoost) [9], which are well-suited to capturing non-linear relationships and handling missing data. Support Vector Machines (SVMs) [10] also remain relevant for high-dimensional classification tasks, particularly when the number of features exceeds the number of observations.

Model evaluation in clinical ML extends beyond standard accuracy metrics and must account for domain-specific constraints, such as class imbalance and asymmetric error costs. Metrics such as the Area Under the Receiver Operating Characteristic curve (AUROC) [11] and the Area Under the Precision-Recall Curve (AUPRC) are widely used, with the latter being particularly informative in scenarios involving rare clinical events [12]. Additionally, calibration has emerged as a critical requirement, ensuring that predicted probabilities correspond to observed event frequencies, thereby supporting reliable clinical decision-making.

Despite these advancements, several methodological challenges persist. Clinical datasets are often characterized by significant class imbalance, where adverse outcomes represent a minority of cases. Techniques such as the Synthetic Minority Over-sampling Technique (SMOTE) [13] and its variants have been proposed to address this issue by generating synthetic samples of the minority class, although their effectiveness may decrease in high-dimensional

settings. Furthermore, missing data, frequently not missing at random, require careful handling through advanced imputation strategies to avoid introducing bias. These challenges, combined with limited sample sizes and heterogeneous feature spaces, underscore the importance of developing models that are both accurate and robust across different clinical environments.

2.2 Data Shift and Generalization in Clinical Machine Learning

The deployment of Machine Learning (ML) models from controlled research environments to real-world clinical settings is frequently challenged by the phenomenon of data shift, defined as a discrepancy between the joint distribution of inputs and labels during training and deployment. In the medical domain, such shifts are typically categorized into three types: covariate shift, prior probability shift, and concept drift.

Covariate shift refers to changes in the distribution of input features $P(x)$ while the conditional relationship $P(y|x)$ remains stable, for example due to variations in patient demographics or hospital referral patterns. Prior probability shift involves changes in the marginal distribution of the outcome $P(y)$, such as an increase in disease prevalence during a pandemic. Concept drift, the most critical form of shift, occurs when the relationship between inputs and outcomes $P(y|x)$ changes over time, often as a result of evolving clinical practices, new treatment protocols, or changes in diagnostic criteria [14, 15].

Generalization under these conditions remains a major challenge in clinical ML. Empirical evidence suggests that models often exhibit significant performance degradation when applied to external datasets, a phenomenon commonly referred to as model decay [16]. This degradation is frequently driven by the reliance on spurious or context-specific correlations that do not generalize across institutions or populations.

The sources of data shift in healthcare are multifactorial. Variability in data acquisition processes, such as differences in imaging protocols or medical devices, can introduce systematic biases. Similarly, heterogeneity in EHR systems and coding practices contributes to inconsistencies in feature representation. Demographic variability across patient populations

further exacerbates these issues, limiting the external validity of predictive models.

To address these challenges, several methodological strategies have been proposed. Domain adaptation techniques aim to align feature distributions between source and target domains, while transfer learning approaches enable the adaptation of pre-trained models to new environments using limited local data. Additionally, robust validation protocols are essential: beyond internal cross-validation, external validation on independent datasets and temporal validation on more recent data are necessary to assess model generalizability.

Calibration techniques, including Platt scaling and isotonic regression [17, 18], are also employed to maintain the reliability of predicted probabilities under distributional changes. Moreover, continuous monitoring and periodic model retraining are increasingly recognized as necessary components of model lifecycle management in clinical settings.

Failure to adequately account for data shift can lead to substantial reductions in model reliability and safety. Studies have documented significant decreases in sensitivity and increases in false positive rates when models are deployed outside their original training context. These findings highlight the importance of explicitly addressing dataset shift during model development and deployment, ensuring that predictive systems capture generalizable clinical relationships rather than institution-specific artifacts.

2.3 Interpretability and Trustworthiness in Clinical Machine Learning

The clinical adoption of Machine Learning (ML) is fundamentally constrained by the trade-off between predictive performance and model interpretability. While complex models such as deep neural networks and gradient-boosted ensembles often achieve superior predictive accuracy, their decision-making processes are typically opaque. In clinical contexts, this lack of transparency raises critical concerns regarding safety, accountability, and trust.

To address this issue, Explainable Artificial Intelligence (XAI) techniques have been developed to provide insight into model behavior. These approaches can be broadly categorized into global and local explanation methods. Global methods, such as permutation feature importance and SHapley Additive Explanations (SHAP), provide an overall view of

feature relevance across the population [19]. SHAP, in particular, is grounded in cooperative game theory and assigns each feature a contribution value based on its marginal impact on model predictions. Local methods, such as Local Interpretable Model-agnostic Explanations (LIME), focus on explaining individual predictions by approximating the model locally with simpler, interpretable models [20, 21].

Beyond interpretability, trustworthiness in clinical ML encompasses additional dimensions, including fairness, reliability, and uncertainty quantification. Bias in training data can lead to systematic disparities in model performance across different patient subgroups, raising important ethical concerns. Fairness criteria, such as equality of opportunity, have been proposed to ensure that predictive performance is consistent across populations, particularly in terms of true positive rates [22].

Uncertainty quantification is another critical component of trustworthy ML systems. In clinical decision-making, it is not sufficient for a model to provide a prediction; it must also convey the degree of confidence associated with that prediction. Approaches such as Bayesian modeling and conformal prediction enable the estimation of predictive uncertainty [23], allowing clinicians to identify cases where model outputs should be interpreted with caution [24].

Regulatory frameworks have further emphasized the importance of transparency and accountability in AI-driven healthcare systems. In particular, the concept of a “right to explanation” has emerged in response to legal requirements [25, 26], highlighting the need for models to provide meaningful and interpretable justifications for their predictions.

Finally, the successful integration of ML systems into clinical workflows depends on usability and alignment with practitioner needs. Even highly accurate and interpretable models may fail in practice if their outputs are not presented in a clear and actionable manner. Therefore, achieving trustworthy clinical ML requires a holistic approach that combines methodological rigor with careful consideration of human factors and real-world constraints.

Chapter 3

Materials and Methods: The Dataset

3.1 Cohort Description

The study population consists of a cohort of 1,710 hospitalized patients who underwent surgical intervention for hip or knee fractures at the *Istituto Ortopedico Galeazzi* in Milan during the 2019 calendar year. The study adopts a retrospective and observational design, including the entire available population without predefined inclusion or exclusion criteria, in order to maximize representativeness with respect to real-world clinical practice (Real World Data).

A summary of the main demographic and clinical characteristics is provided in Table 3.1.

Descriptive analysis highlights a demographic profile consistent with elderly orthopedic populations. The mean age is 69.4 years (SD: 10.73), with a median of 70 years and an interquartile range (IQR) of 15 years. As shown in Figure 3.1, the age distribution is concentrated in the 60–75 year range.

Table 3.1: **Table 1 – Summary of Principal Demographic and Clinical Characteristics**

Characteristic	**Value / Distribution**
Total number of patients	1,710
Mean age (±SD)	69.4 ± 10.73 years
Median (IQR)	70 (IQR: 15)
Female gender	62.34%
Male gender	36.73%
Not specified	0.93%
Most common comorbidities	Hypertension (57.8%), Type 2 Diabetes (11.2%), Thyroid disease (13.9%), Dyslipidemia (7.4%)
Primary occupation	Retired (59.3%)
Prevalent living condition	Resides with family (81.3%)

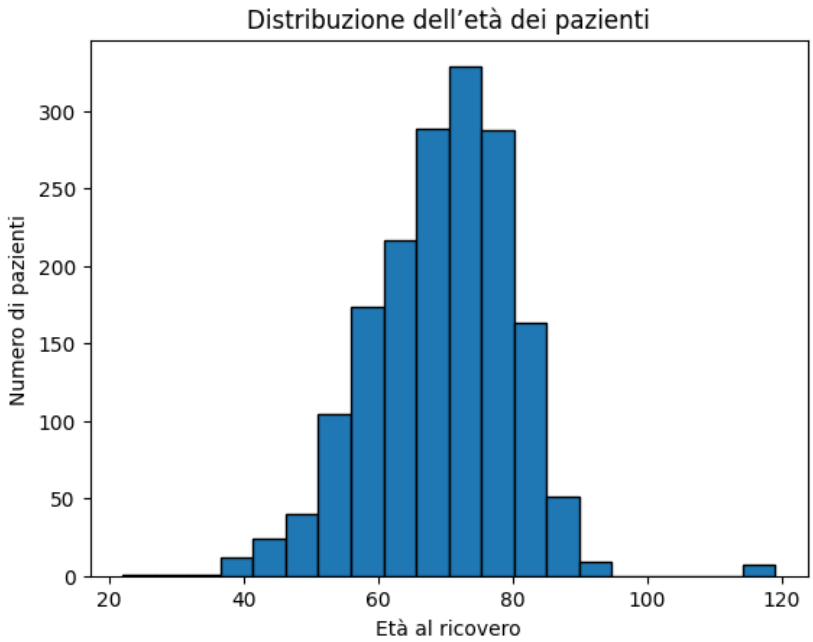

Figure 3.1: Patient age distribution.

Gender distribution shows a marked predominance of female patients (62.34%, $n \approx 1{,}066$) compared to males (36.73%, $n \approx 628$), with 0.93% of records presenting ambiguous coding (see Figure 3.2).

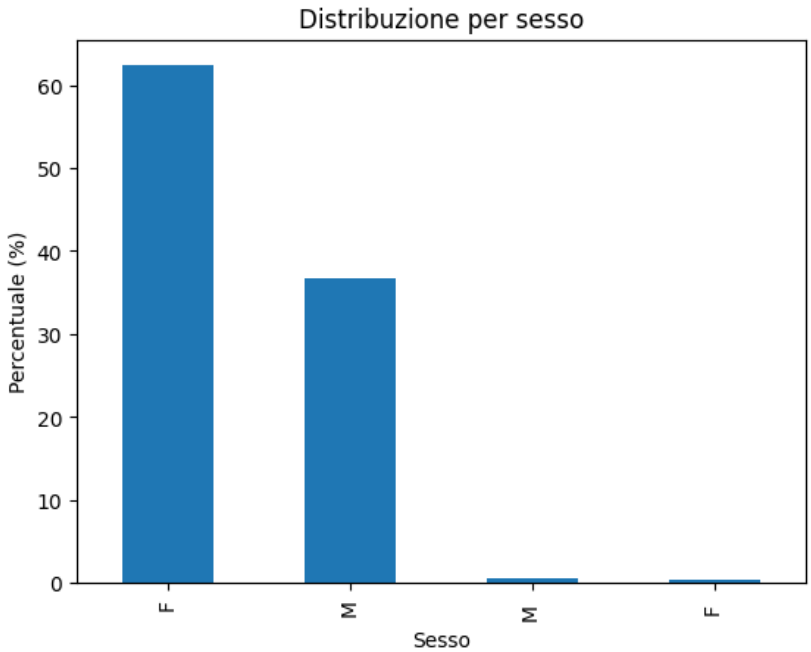

Figure 3.2: Patient sex distribution.

From a clinical perspective, the cohort is characterized by a substantial burden of comorbidities, particularly cardiovascular and metabolic conditions. The most prevalent include arterial hypertension (57.8%, $n = 988$), thyroid diseases (13.9%, $n = 237$), cardiovascular diseases excluding hypertension (11.6%, $n = 199$), and type 2 diabetes mellitus (11.2%, $n = 192$). Functional status at admission is quantified by a mean Barthel Index (BI) of 70.28 (SD: 12.01) and a mean Functional Independence Measure (FIM) score of 96.66 (SD: 9.16). Additionally, the mean anesthesiological risk class (ASA) is 1.95 (SD: 0.35).

Socio-economic and housing variables are incorporated as potential predictors of discharge outcomes. Specifically, 59.3% of patients are retired, and 13.9% are classified as homemakers. Regarding living conditions, 81.3% of patients reside with family members, while 17.3% live alone, predominantly individuals over 65 years of age. This variable demonstrates a statistically significant negative correlation with the social outcome ($r = -0.29$, $p < 0.05$), supporting its inclusion to improve the specificity of classification models.

Prior to modeling, the dataset undergoes a series of statistical quality and robustness assessments. Outlier detection is performed using the interquartile range rule ($Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$), identifying the highest prevalence of outliers in admission pain (13.16%) and length of stay (7.95%). These observations are retained to preserve intrinsic clinical variability, which is essential for the generalization capacity of ensemble models. Missing data analysis reveals a substantial proportion of missing values in the *pharmacological therapy* variable (21.81%), addressed through multiple imputation using the MICE approach (IterativeImputer in `scikit-learn`), fitted exclusively on the training subset (80% of the data, n = 1,257) to prevent data leakage.

Furthermore, a temporal data shift is identified through comparison with an independent cohort from 2018 (n = 360). Kolmogorov–Smirnov tests indicate statistically significant differences (p = 0.0) for variables such as surgical duration and length of stay. This finding is corroborated by adversarial validation, in which a Random Forest classifier achieves an accuracy of 1.0 in distinguishing between the 2018 and 2019 cohorts, confirming the presence of distinct underlying distributions.

In summary, the analyzed cohort provides a multidimensional dataset comprising 56 initial features, expanded to 77 after preprocessing, reflecting the complexity of orthopedic clinical practice. This complexity introduces relevant methodological challenges, particularly related to non-linear relationships and temporal distributional shifts.

3.2 Target Definition (Gold Standard)

The definition of the Gold Standard constitutes a critical phase of the project, as it determines the dependent variables that Machine Learning models are required to learn. In the present study, clinical outcomes are organized into three hierarchical categories—primary, secondary, and exploratory—based on their relevance to hospital management and functional recovery.

The primary outcome is the duration of post-operative stay in the rehabilitation unit, modeled as a continuous variable within a regression framework. As summarized in Table 3.2, this target exhibits a mean of 10.28 days (SD: 3.72), a median of 10 days, and a range between 1 and 34 days (IQR: 5).

Table 3.2: **Table 2a – Continuous Targets (Regression)**

Outcome	**Mean**	**Std. Dev. / Range / IQR**
Rehabilitation stay duration	10.28 days	SD 3.72; Min 1 – Max 34; IQR 5
BI_post_int (Barthel Index)	95.71	SD 6.30; Min 27 – Max 108; IQR 4
FIM_post_int (Functional Independence Measure)	116.48	SD 6.07; Min 59 – Max 129; IQR 6

Data quality is notably high, with a missing rate of only 0.06%, ensuring reliability for model training. The choice of a continuous formulation is motivated by the need to provide precise prognostic estimates, supporting both bed management optimization and individualized care pathway planning.

Secondary outcomes include both categorical and continuous variables, detailed in Table 3.3. The social outcome, defined as discharge destination, was initially encoded into three categories: discharge to home with relatives (90.6%), discharge to home alone (5.9%), and discharge to an institution (3.5%). Due to the pronounced class imbalance, a binarization strategy was adopted, aggregating the two minority classes to improve classification performance and facilitate the identification of patients at risk of reduced residential autonomy. Similarly, the care burden variable, representing required assistance hours, was binarized to mitigate distributional bias, with 76.4% of patients requiring less than three hours of assistance and 23.2% requiring none.

Functional outcomes are modeled through regression tasks involving the post-operative Barthel Index (BI) and Functional Independence Measure (FIM). The BI exhibits a mean of 95.71 (SD: 6.30) with a range of 27–108, while the FIM shows a mean of 116.48 (SD: 6.07) with a range of 59–129. Despite a strong correlation between the two indices ($r = 0.83$), both

Table 3.3: **Table 2b – Categorical Targets (Classification)**

Outcome	**Distribution / Percentage**
Social outcome (discharge location)	Home with relatives 90.6%; Home alone 5.9%; Institution 3.5%; NaN 0.1%
Care burden	None 23.2%; <3h 76.4%; <6h 0.2%; Day and night 0.1%; NaN 0.1%
Ambulation	Autonomous 93.9%; Autonomous <50m 4.2%; With supervisor 1.7%; Impossible 0.2%; NaN 0.1%
Walking aids	2 Canadian crutches 98.3%; Walker 1.2%; None 0.2%; Wheelchair 0.1%; 1 Cane 0.1%; NaN 0.1%

targets are retained to capture complementary aspects of functional independence.

Exploratory outcomes include additional variables related to motor autonomy. Specifically, 93.9% of patients are autonomous in ambulation at discharge, while 98.3% utilize two Canadian crutches, indicating very low variance for this feature. These variables are considered optional targets due to their limited discriminative power.

A critical limitation concerns the quality and origin of the labels. The annotation process is not documented, and no information is available regarding inter-rater agreement or conflict resolution strategies (e.g., majority voting). This limitation must be considered when interpreting model performance, as label quality directly affects generalization capability. Furthermore, the analysis of data shift adversarial validation—implemented via a Random Forest classifier distinguishing training and test sets—achieves an accuracy of 1.0, indicating substantial temporal variation not only in feature distributions but also in target relationships (concept drift). Consequently, model recalibration is required to ensure reliable deployment in clinical settings.

3.3 Feature Engineering and Data Preprocessing

The feature engineering phase transformed the initial raw dataset, composed of 56 independent variables, into a structured and standardized set of predictors suitable for model training. At the conclusion of this process, the feature space was expanded to 77 variables. The entire workflow was designed with rigorous methodological criteria to ensure numerical stability and to prevent data leakage.

Categorical variables were processed through two complementary encoding strategies. First, domain-informed binary features were introduced to simplify dimensionality and capture clinically relevant information. These include variables such as *Sesso_F* (gender binarization with female as the positive class), *Rischio_bio_bin*, *Trasfusione_bin*, and *Farmaci_extra_bin*, representing the presence of biological risk, transfusion requirement, and additional pharmacological treatments, respectively. Additionally, a derived variable for sedentary status was constructed by aggregating occupational categories associated with reduced physical activity. Second, nominal variables with more than two levels—such as diagnostic and procedural ICD codes, anesthesia type, ADL scores, housing conditions, and prior prostheses—were encoded using one-hot encoding using the `scikit-learn` library. To avoid perfect multicollinearity (dummy variable trap), the parameter `drop_first=True` was applied, ensuring the stability of linear models such as Ridge and Logistic Regression. Furthermore, the variables representing pre-existing conditions (1–9) were expanded into a sparse binary matrix, where each column indicates the presence or absence of a specific comorbidity.

Missing data handling was addressed through an advanced imputation strategy. The variable *pharmacological therapy*, characterized by a missing rate of 21.81%, was imputed using Multiple Imputation by Chained Equations (MICE), implemented via the `IterativeImputer` in `scikit-learn`. To prevent data leakage, the imputer was fitted exclusively on the training set (1,257 patients) and subsequently applied to the validation set (315 patients). Post-processing included rounding the imputed continuous values to the original categorical levels (0, 0.5, 1) to preserve data consistency. Empirical evaluation confirmed the effectiveness of this approach: in the SVR model, the imputed dataset achieved an R^2 of 0.265, compared to 0.216 obtained under a complete-case analysis.

Continuous variables—including age, pre-intervention BMI, length of stay, surgical duration, BI, FIM, and hemoglobin at admission—exhibited heterogeneous scales and units. Consequently, Z-score standardization was applied using the `StandardScaler`, transforming each variable to have mean $\mu = 0$ and standard deviation $\sigma = 1$. This step is essential for distance-based algorithms, such as Support Vector Machines (SVM) and Multi-Layer Perceptrons (MLP), as well as for regularized models (Ridge and Lasso), preventing variables with larger numerical ranges (e.g., surgical duration, with a maximum of 386 minutes) from disproportionately influencing model training.

Additional preprocessing steps were implemented to address class imbalance and data shift. Target variables such as care burden and social outcome were binarized due to strong imbalance (e.g., 90.6% of patients discharged home with relatives), aggregating minority classes to improve classification performance. To mitigate covariate shift identified between cohorts—supported by Kolmogorov–Smirnov tests with $p = 0.0$ for duration-related variables—importance weighting was explored. Specifically, observation weights were derived from a discriminative classifier estimating the probability of belonging to the external test set relative to the training set. To ensure numerical stability, weights were capped at the 95th percentile, preventing extreme values from dominating the optimization process.

The integration of these preprocessing techniques resulted in finalized datasets comprising 1,257 samples for training and 315 for validation, with a total of 77 engineered features, providing a robust foundation for subsequent modeling.

Chapter 4

Data Preparation and Preprocessing

4.1 Data Cleaning and Data Quality Analysis

The data cleaning and quality assessment phase was essential to transform raw data extracted from the hospital Electronic Health Records (EHR) of the *Istituto Ortopedico Galeazzi* into a robust dataset suitable for modeling. Although the original dataset of 1,710 patients demonstrated strong representativeness, several issues were identified in terms of completeness and format consistency, requiring systematic preprocessing.

No duplicate records were detected ($n = 0$). However, substantial heterogeneity was observed in textual and categorical variables (e.g., gender, occupation, anesthesia type, ADL scores, and pre-existing conditions), characterized by inconsistent capitalization, redundant labels, and superfluous whitespace. Consequently, a standardized normalization procedure was applied, converting entries into Title-Case or Upper-Case formats. Semantically equivalent categories (e.g., variants of the label "Casalinga") were consolidated to reduce noise and unnecessary dimensionality. Additionally, variables corresponding to ICD diagnostic and procedural codes, as well as ASA classification, were explicitly cast as string types to ensure correct handling during subsequent encoding phases.

Logical consistency checks were implemented to identify implausible values and potential data entry errors. Specifically, surgical durations below 20 minutes were considered unrealistic and recoded as missing values (NaN). Similarly, to prevent distortion in regression tasks, post-operative Barthel Index (BI_post_int) values greater than 100 and Functional Independence Measure (FIM_post_int) values exceeding 126 were invalidated and set to NaN. Laboratory

variables were also standardized; for example, hemoglobin at admission (Hb ingresso) was converted from comma-based decimal notation to floating-point format.

Outlier detection was performed using the Interquartile Range (IQR) method, defining anomalies as values outside the interval $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$. The highest proportions of outliers were observed in admission pain (VNS) (13.16%), pre-operative length of stay (7.95%), post-operative Barthel Index (6.49%), and surgical duration (3.22%). Despite this, no automatic removal or capping procedures were applied. This decision was motivated by the need to preserve the intrinsic variability of the clinical population, including complex or atypical cases, which is critical for the generalization capability of ensemble models.

The missing data pattern was heterogeneous and non-random. Variables such as "Pre-existing conditions 1–9" exhibited extremely high missingness (up to 99.94%), which was handled by introducing an explicit "Missing" category to retain information about non-observation. Variables with low missing rates (e.g., admission pain at 4.39% and pre-intervention BMI at 1.29%) were handled through case-wise deletion, minimizing the impact on statistical power. For the variable *pharmacological therapy*, characterized by 21.81% missingness, multiple imputation via MICE (IterativeImputer) was employed, fitted exclusively on the training set to prevent data leakage. This approach led to measurable improvements in model performance, with the R^2 of the SVR model increasing from 0.21 to 0.265.

Finally, temporal data quality was assessed through a comparison between the 2019 training cohort and an external 2018 cohort. A permutation test yielded a p-value of 0.0, indicating a critical distributional difference. This finding was reinforced by adversarial validation, where a Random Forest classifier achieved an accuracy of 1.0 in distinguishing between the two cohorts. Furthermore, Kolmogorov–Smirnov tests identified significant deviations ($p = 0.0$) in variables such as surgical duration and length of stay, suggesting a degradation in generalization performance attributable to temporal changes in clinical processes or patient composition.

4.2 Correlation Analysis and Outlier Identification

This phase of the analysis aimed to quantify linear relationships among clinical variables and assess the integrity of the signal contained in the dataset, while simultaneously identifying observations that significantly deviate from the underlying distributions.

Outlier detection was performed on key numerical variables relevant for modeling, including age, BMI, surgical duration, Barthel Index (BI), Functional Independence Measure (FIM), hemoglobin at admission, and pain score (VNS). The Interquartile Range (IQR) method was applied, defining outliers as values outside the interval $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$. The highest proportions of anomalous values were observed for admission pain (VNS) (13.16%), pre-operative length of stay (7.95%), post-operative BI (6.49%), surgical duration (3.22%), post-operative FIM (2.87%), and rehabilitation length of stay (2.16%). Despite these values, no automatic correction procedures—such as removal or capping—were applied. This decision reflects the interpretation of these observations as representative of genuine variability in patient conditions and procedural characteristics, rather than measurement errors, thereby preserving the dataset's ability to support generalization, particularly for ensemble models.

To investigate redundancy and interdependencies among predictors, a correlation matrix and corresponding heatmap were generated, adopting a threshold of $|\text{corr}| > 0.7$ to identify strong associations. The most significant correlation was observed between BI and FIM ($r = 0.83$). Despite this high collinearity, both variables were retained in the feature set. This choice is motivated by their complementary nature, as they may capture distinct aspects of functional status, thereby potentially enhancing the explanatory power of the models.

Further analysis focused on the linear relationships between predictors and target variables. For rehabilitation length of stay, surgical duration exhibited the strongest positive correlation ($r = 0.35$, $p < 0.05$), while the midvastus surgical approach showed the most pronounced negative correlation ($r = -0.28$). Regarding the binarized social outcome, living alone (*Vive con_Da solo*) was positively correlated with the risk of non-protected discharge ($r = 0.29$), whereas living with family members (*Vive con_Familiari*) demonstrated a negative association ($r = -0.28$). For the post-operative Barthel Index, the strongest predictors were the pre-operative BI ($r = 0.49$) and pre-operative FIM ($r = 0.43$), while non-autonomy in postural

transitions showed a substantial negative correlation ($r = -0.39$).

Finally, to assess model generalization, a data shift analysis was conducted using an external cohort from 2018. Adversarial validation, implemented via a Random Forest classifier trained to distinguish between the 2019 training set and the 2018 test set, achieved an accuracy of 1.0, indicating perfectly separable distributions. Complementary Kolmogorov–Smirnov tests identified statistically significant differences ($p = 0.0$) specifically for surgical duration and length of stay. These results suggest the presence of temporal distributional shifts, potentially associated with changes in clinical processes or patient composition, which directly impact model generalization.

4.3 Missing Data Management

The handling of missing data represents a critical component of the preprocessing pipeline, aimed at balancing the preservation of clinically relevant information with the reduction of potential algorithmic bias. The dataset exhibits a heterogeneous missingness pattern, with specific variables requiring differentiated treatment strategies.

A quantitative assessment of the 56 independent features allowed the identification of two primary categories of missingness. Variables with low incidence of missing values (less than 5%) include admission pain (VNS), with 4.39% missingness, and pre-intervention BMI, with 1.29%. For these predictors, a complete-case analysis was adopted, as the removal of a limited number of observations does not significantly compromise statistical power. In contrast, several variables exhibit high missingness rates (greater than 10%). In particular, the set of variables representing pre-existing conditions (1–9) shows missing rates ranging from 21.64% (Condition 1) up to 99.94% (Conditions 8 and 9). Additionally, the *pharmacological therapy* variable presents a missing rate of 21.81%.

For variables with substantial missingness, deletion strategies were avoided to prevent the loss of clinically informative patterns. Instead, two complementary approaches were implemented. First, for the pre-existing conditions variables, missing values were encoded as an explicit “Missing” category. This approach preserves the information associated with non-observation without introducing arbitrary estimates, effectively transforming missingness into

a potentially informative feature. Second, for the *pharmacological therapy* variable, multiple imputation via MICE (Multiple Imputation by Chained Equations) was applied using the `IterativeImputer` implementation in `scikit-learn`. To prevent data leakage, the imputation model was fitted exclusively on the training subset (80% of the data) and subsequently applied to the validation subset (20%). Furthermore, post-imputation values were rounded to the original categorical levels (0, 0.5, 1) to maintain consistency and interpretability.

The impact of the imputation strategy was systematically evaluated through comparative experiments, training models on both imputed and non-imputed datasets. The results demonstrate a consistent improvement in predictive performance across all models, with reductions in Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), accompanied by increases in the coefficient of determination (R^2). In the primary regression task, the Support Vector Regressor (SVR) showed the most significant improvement, achieving an R^2 of 0.265 with imputation compared to 0.216 without (an increase of approximately +0.05). For the prediction of post-operative Barthel Index, linear models (Ridge and Lasso) exhibited moderate improvements, with MAE decreasing from approximately 2.92 to a range of 2.69–2.75. In contrast, tree-based models (Random Forest and LightGBM) demonstrated inherent robustness to missing data, with only marginal performance variations between the two scenarios.

In summary, the adoption of multiple imputation, combined with explicit categorical encoding for highly incomplete variables, effectively mitigates the impact of missing data, enhancing both the stability and consistency of model predictions, particularly for flexible algorithms such as SVR and LightGBM.

4.4 Preprocessing Pipeline

The preprocessing pipeline constitutes the methodological core of the data preparation process, designed to transform raw clinical records from the Galeazzi Institute into a structured and standardized dataset suitable for predictive modeling. The workflow was explicitly developed to ensure numerical stability, prevent data leakage, and address the intrinsic challenges of Real World Data. Starting from an initial set of 56 independent variables, the pipeline

systematically expanded the feature space to 77 predictors through sequential and controlled transformations.

The initial phase focused on data cleaning and sanity checks to ensure internal consistency. Categorical variables, including sex, profession, and anesthesia type, were standardized by removing extraneous whitespace and harmonizing capitalization using title-case or upper-case formats. Redundant labels were consolidated to reduce semantic noise and dimensionality. Furthermore, clinically implausible values were identified and treated as missing, specifically intervention durations below 20 minutes and out-of-range functional scores (Barthel Index > 100, Functional Independence Measure > 126). Variable types were explicitly cast to ensure correct downstream processing: ICD codes and ASA class were converted to string format, while laboratory variables such as hemoglobin at admission were cast to numeric format.

Subsequently, missing data were addressed through a differentiated strategy based on their prevalence. For the variable "Pharmacological Therapy," which exhibited a missing rate of 21.81%, multiple imputation via chained equations (MICE) was implemented using the `IterativeImputer` from `scikit-learn`. To prevent data leakage, the imputation model was fitted exclusively on the training set (80%) and subsequently applied to the validation set (20%). Imputed values were post-processed by rounding them to the original categorical levels (0, 0.5, 1), thereby preserving interpretability. This approach yielded measurable improvements in predictive performance; for instance, the Support Vector Regressor achieved an increase in R^2 from 0.216 to 0.265. In contrast, variables with missing rates below 5% were handled via complete-case analysis.

Feature transformation and encoding were then applied to accommodate the heterogeneous nature of the dataset. Binary indicators were engineered for clinically relevant variables (e.g., `Sesso_F`, `Trasfusione_bin`, `Rischio_bio_bin`, and a derived `Sedentarieta` variable). Nominal variables with multiple categories, including ICD codes, anesthesia type, living conditions, and assistive devices, were encoded using one-hot encoding with `drop_first=True` to avoid perfect multicollinearity. Additionally, the set of pre-existing conditions was expanded into a sparse binary matrix representing the presence or absence of each condition. Continuous variables (age, BMI, hemoglobin at admission, Barthel Index, Functional Inde-

pendence Measure, and hospitalization durations) were standardized using Z-score normalization ($\mu = 0, \sigma = 1$), a necessary step for distance-based algorithms such as Support Vector Machines and Multi-Layer Perceptrons.

Finally, the pipeline incorporated strategies to address class imbalance and temporal data shift. Given the strong imbalance in outcomes (e.g., 90.6% of patients discharged home with relatives), the "Social Outcome" and "Care Burden" variables were binarized by aggregating minority classes. Data shift was formally assessed through adversarial validation, where a Random Forest classifier achieved an accuracy of 1.0 in distinguishing between the 2019 training cohort and the 2018 external test set. This finding was further supported by Kolmogorov–Smirnov tests indicating significant distributional differences ($p = 0.0$) in duration-related variables. To mitigate covariate shift, an importance weighting strategy was explored, with weights computed as $w(x) = p_{OS}(x)/p_{IS}(x)$ and capped at the 95th percentile to prevent numerical instability.

At the conclusion of the preprocessing pipeline, the dataset was partitioned into a training set comprising 1,257 patients and a validation set of 315 patients, each described by 77 features. This final configuration ensures a consistent, numerically stable, and clinically interpretable input space, suitable for subsequent model training and external validation.

Chapter 5

Experimental Setup and Modeling

5.1 Model Architectures

The modeling strategy adopted in this study encompasses a diverse set of Machine Learning and Deep Learning algorithms, implemented using `scikit-learn`, `LightGBM`, `Keras/TensorFlow`, and the `scikeras` wrapper. This architectural diversification was intentionally designed to compare highly interpretable linear models with more complex non-linear approaches capable of capturing intricate interactions among the 77 clinical and functional features.

Linear models were initially employed for both regression tasks (rehabilitation length of stay and post-operative Barthel Index) and classification tasks (Social Outcome), due to their analytical transparency. Specifically, Ridge and Lasso regression were used to address multicollinearity, particularly relevant given the strong correlation between Barthel Index and Functional Independence Measure ($r = 0.83$), through L2 and L1 regularization, respectively. The regularization parameter α was optimized using `GridSearchCV` with 5-fold cross-validation. Quantitatively, the optimal Ridge model ($\alpha = 100$) achieved an R^2 of 0.23 in cross-validation for rehabilitation duration, while Lasso ($\alpha = 0.1$) reached the same performance. Logistic Regression was applied to the binary classification of the Social Outcome; to address class imbalance (90.6% majority class), it was integrated into pipelines incorporating either synthetic oversampling (SMOTENC) or class reweighting (`class_weight='balanced'`).

Furthermore, in alignment with preliminary analyses indicating weak linear associations (e.g., $r = 0.35$ between intervention duration and hospitalization length), tree-based ensemble methods were introduced. Random Forest, based on bagging, was optimized through

`GridSearchCV` (e.g., `n_estimators` in the range 100–500 and `max_depth` up to 30) and demonstrated statistically significant superiority over linear models ($p_{\text{corr}} < 0.001$, Wilcoxon test). LightGBM, a gradient boosting framework with leaf-wise growth, was selected for its computational efficiency. Class imbalance was handled using the native `scale_pos_weight` parameter, computed as the ratio of negative to positive samples. This configuration yielded the best classification performance for the Social Outcome, with an F_2-score of 0.656 and a Balanced Accuracy of 0.856.

In addition, kernel-based and neural network models were evaluated to further explore non-linear relationships. Support Vector Machines (SVR/SVC), leveraging margin maximization and kernel transformations, were configured with an RBF kernel. The optimal regression configuration ($C = 1$, $\epsilon = 1$) achieved the highest validation performance among all tested models, with an R^2 of 0.265. A Multi-Layer Perceptron (MLP), composed of dense layers with ReLU activation, dropout regularization (up to 0.3), and Adam/RMSprop optimizers, was also implemented. However, despite its flexibility, the MLP exhibited limited generalization on relatively small sample sizes, yielding a negative R^2 for the post-operative Barthel Index.

All architectures were evaluated under a rigorous experimental protocol. Models were trained and tested on both complete-case datasets and datasets enriched using MICE imputation, revealing that non-linear models (SVR, Random Forest, LightGBM) benefited more substantially from increased data completeness. Performance differences, measured by MAE, RMSE, and R^2, were statistically assessed using the Wilcoxon signed-rank test (paired comparisons) and the Mann–Whitney U test (independent samples). The Benjamini–Hochberg false discovery rate (FDR) correction was systematically applied to control for multiple comparisons, confirming that ensemble models significantly outperformed linear approaches ($p_{\text{corr}} < 0.05$). Finally, environmental sustainability was monitored, with training energy consumption estimated at approximately 0.005 kg CO_2eq on an Intel i7 CPU for the most efficient models, such as LightGBM.

5.2 Experimental Design and Validation Protocol

The experimental design was structured to ensure a rigorous evaluation of model robustness and sensitivity under varying data conditions, with particular emphasis on missing data handling and generalization capability on independent cohorts. The protocol was designed to systematically assess both internal performance and external validity, while minimizing sources of bias such as data leakage.

The original dataset (2019 cohort, $n = 1{,}710$) was partitioned into two independent subsets: a training set (80%, $n = 1{,}257$) used for model fitting and hyperparameter optimization, and a validation set (20%, $n = 315$) reserved exclusively for internal performance evaluation. The split was performed using stratified sampling with respect to patient sex, ensuring that the gender distribution (62.34% female) was consistently preserved across both subsets. This strategy was adopted to prevent sampling bias and to guarantee that preprocessing steps, including scaling and imputation, were applied without introducing data leakage.

Furthermore, an ablation study was conducted to evaluate the impact of missing data treatment, focusing on the "Pharmacological Therapy" variable (21.81% missing). Two experimental scenarios were defined: (i) a complete-case analysis excluding observations with missing values, and (ii) an imputed dataset using Multiple Imputation by Chained Equations (MICE) implemented via `IterativeImputer`. This comparison enabled a direct quantification of the imputation effect; for instance, in the rehabilitation length-of-stay task, the Support Vector Regressor improved its R^2 from 0.216 (complete-case) to 0.265 (with imputation).

Hyperparameter optimization was conducted separately for each model architecture using `GridSearchCV` and `RandomizedSearchCV`, combined with 5-fold cross-validation (reduced to 3-fold for Multi-Layer Perceptron models due to computational constraints). Optimization metrics were selected according to the task: the coefficient of determination (R^2) for regression, and F_2-score alongside Balanced Accuracy for classification. The preference for the F_2-score over the F_1-score reflects a methodological choice to prioritize recall, thereby reducing the incidence of false negatives in clinically relevant classification tasks.

Model comparison and statistical validation were conducted using a non-parametric frame-

work. Specifically, the Wilcoxon signed-rank test was applied to paired samples (imputed scenario) based on absolute residuals, while the Mann–Whitney U test was used for independent samples (complete-case scenario), where prediction sets differed in size due to missing value removal. To control for multiple hypothesis testing, the Benjamini–Hochberg false discovery rate (FDR) correction was systematically applied, with statistical significance established at $p_{\text{corr}} < 0.05$.

Finally, external validation was performed on an independent cohort collected in 2018 ($n = 360$) to assess model generalization. Adversarial validation, implemented using a Random Forest classifier trained to distinguish between 2019 (label 0) and 2018 (label 1) data, achieved an accuracy of 1.0, indicating a substantial distributional shift. This finding was corroborated by Kolmogorov–Smirnov tests, which identified significant differences ($p = 0.0$) in the distributions of "Intervention Duration" and "Length of Stay." To quantify uncertainty in performance estimates, 95% confidence intervals for evaluation metrics (e.g., MAE, R^2) were computed using bootstrapping with 1,000 resamples.

Overall, this experimental framework ensures a robust and reproducible evaluation of predictive performance, explicitly accounting for missing data mechanisms and temporal distribution shifts.

Chapter 6

Internal Validation and Results

6.1 Model Performance Evaluation

This section provides a comprehensive evaluation of the predictive models, stratified by clinical targets and methodological configurations. In particular, the analysis quantifies the impact of key preprocessing choices, such as missing data imputation, on internal generalization performance. Model evaluation was conducted using error-based metrics and coefficients of determination, with 95% confidence intervals estimated via bootstrapping (n = 1000).

6.1.1 Primary Outcome: Length of Stay

The prediction of rehabilitation length of stay was assessed by comparing linear and non-linear models on the validation set. In alignment with the experimental design, the application of multiple imputation via the IterativeImputer (MICE) enabled the use of the full dataset, yielding improved predictive performance. Significant correlations with the dependent variable are illustrated in Figure 6.1.

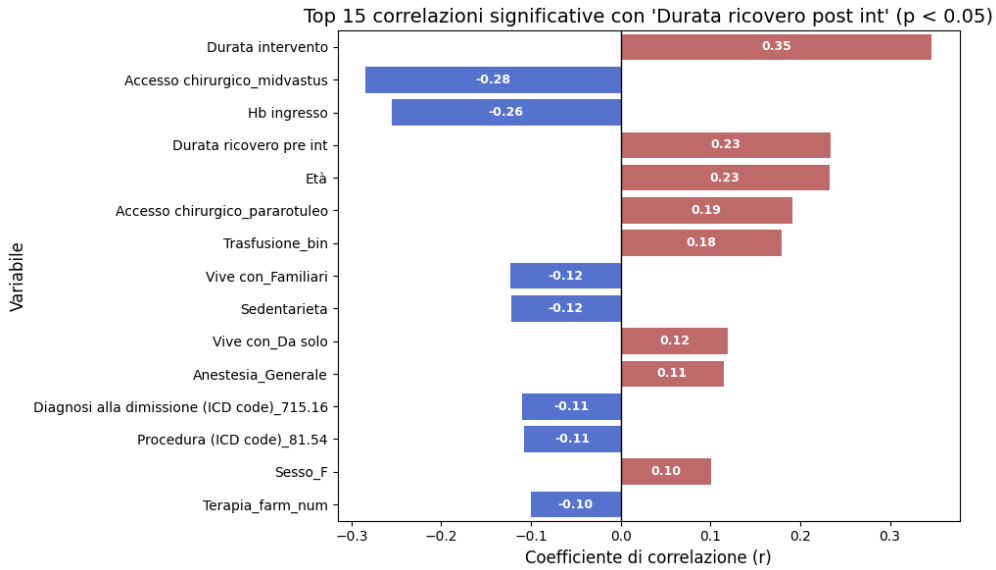

Figure 6.1: Significant correlations with the dependent variable "Rehabilitation Stay Duration."

Among all models, the optimized Support Vector Regressor (SVR) achieved the best overall performance, with an $R^2 = 0.265$ (95% CI: [0.187, 0.346]) and a Mean Absolute Error (MAE) of 2.26 days. Similarly, the optimized LightGBM model demonstrated competitive results, with $R^2 = 0.250$ and the lowest MAE (2.20 days). Detailed performance metrics for all models are presented in Table 6.1.

Table 6.1: Model performance on the Validation set (with imputation)

Model	**MAE [95% CI]**	**RMSE [95% CI]**	**R^2 [95% CI]**	**MAE/SD**	**RMSE/SD**
Ridge Regression ($\alpha = 100$)	2.43 [2.20, 2.71]	3.31 [2.77, 3.92]	0.210 [0.131, 0.288]	0.654	0.889
Lasso Regression ($\alpha = 0.1$)	2.48 [2.24, 2.77]	3.36 [2.81, 3.99]	0.185 [0.104, 0.263]	0.667	0.903
Random Forest (optimized)	2.22 [1.95, 2.48]	3.25 [2.70, 3.88]	0.237 [0.126, 0.333]	0.596	0.874
LightGBM (optimized)	2.20 [1.94, 2.47]	3.22 [2.66, 3.86]	0.250 [0.144, 0.350]	0.591	0.866
Decision Tree (optimized)	2.32 [2.03, 2.61]	3.47 [2.90, 4.10]	0.131 [-0.025, 0.253]	0.623	0.932
SVR (optimized)	2.26 [2.02, 2.51]	3.19 [2.65, 3.80]	0.265 [0.187, 0.346]	0.607	0.857
MLP (optimized)	2.29 [2.05, 2.55]	3.19 [2.73, 3.71]	0.264 [0.145, 0.348]	0.615	0.858

Conversely, linear models exhibited inferior performance: Ridge Regression achieved $R^2 = 0.210$, while Lasso Regression reached $R^2 = 0.185$. This discrepancy suggests the

presence of non-linear relationships within the clinical data that are not adequately captured by linear formulations. To visualize the relative accuracy and correlation of the different approaches, a Taylor Diagram is provided in Figure 6.2.

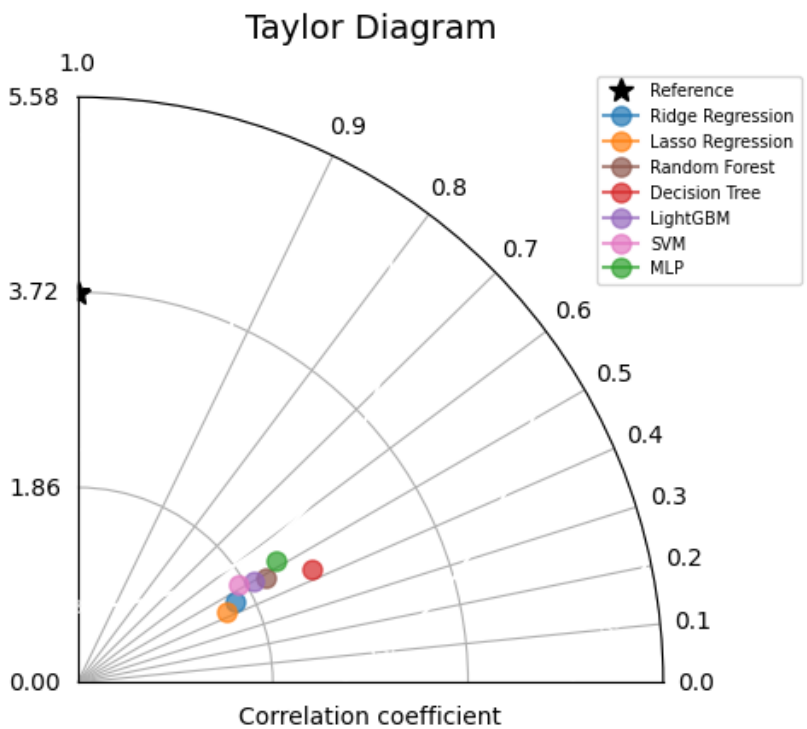

Figure 6.2: Taylor Diagram for model comparison.

To validate these findings, a Wilcoxon signed-rank test was applied to the distribution of absolute errors. Furthermore, the Benjamini–Hochberg correction was used to control the false discovery rate. The results indicate that differences between linear (Ridge, Lasso) and non-linear models (Random Forest, LightGBM, SVR) are statistically significant ($p_{\text{corr}} < 0.05$), whereas differences within the set of non-linear models are not significant. In the absence of imputation, comparisons were conducted using the Mann–Whitney U test due to unequal prediction lengths; in this setting, imputation improved SVR performance by approximately +0.05 in R^2 compared to the complete-case analysis.

6.1.2 Secondary Outcome: Social Discharge Classification

The classification of discharge destination required evaluation metrics robust to severe class imbalance (90.6% majority class). Consequently, the F2-score was adopted as the primary metric, prioritizing recall over precision in order to minimize false negatives. Significant predictors and correlations associated with discharge location are shown in Figure 6.3.

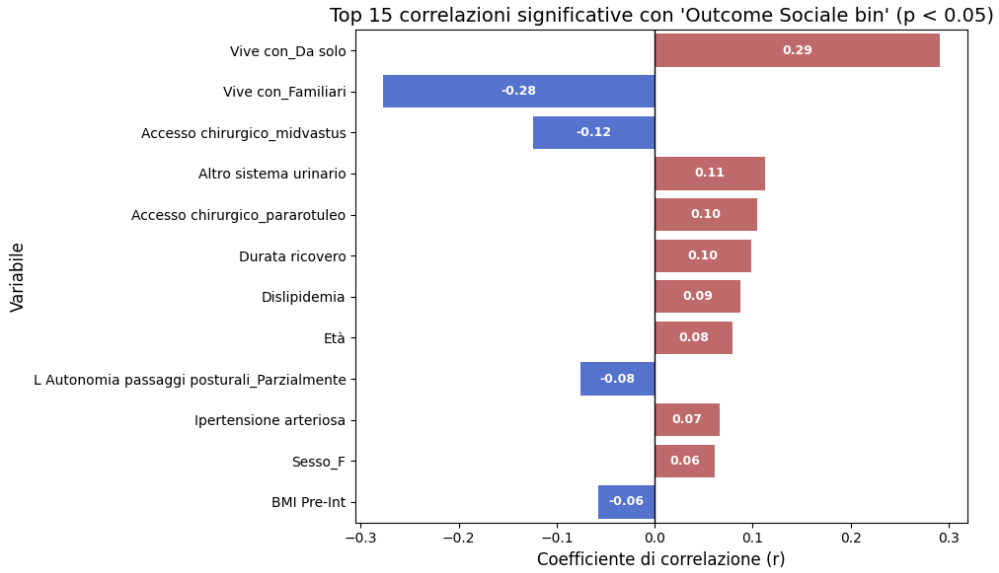

Figure 6.3: Significant correlations with the dependent variable "Discharge Location."

The LightGBM model with class weighting (`scale_pos_weight`) achieved the best performance, with an F2-score of 0.656 (95% CI: [0.467, 0.785]) and a Balanced Accuracy of 0.856. Detailed performance metrics for all tested classification models are presented in Table 6.2.

Table 6.2: Model performance on the validation set for discharge location classification

Model	**F2-score [95% CI]**	**Balanced Accuracy [95% CI]**
SMOTENC + Logistic	0.576 [0.392, 0.710]	0.827 [0.719, 0.909]
Balanced Random Forest	0.635 [0.444, 0.769]	0.849 [0.742, 0.932]
LightGBM + scale_pos_weight	0.656 [0.467, 0.785]	0.856 [0.748, 0.937]
BalancedBaggingClassifier + Decision Tree	0.584 [0.402, 0.720]	0.831 [0.721, 0.912]
NearMiss v3 + SVC	0.497 [0.343, 0.615]	0.809 [0.725, 0.870]
NearMiss v3 + MLP	0.525 [0.369, 0.639]	0.836 [0.768, 0.880]

The Balanced Random Forest also demonstrated strong performance (F2 = 0.635), confirming the effectiveness of ensemble-based approaches in imbalanced settings. In contrast, undersampling strategies such as NearMiss v3 led to reduced performance (F2 = 0.497 for SVC and 0.525 for MLP), indicating potential information loss. A visual comparison of these metrics with their respective confidence intervals is provided in Figure 6.4.

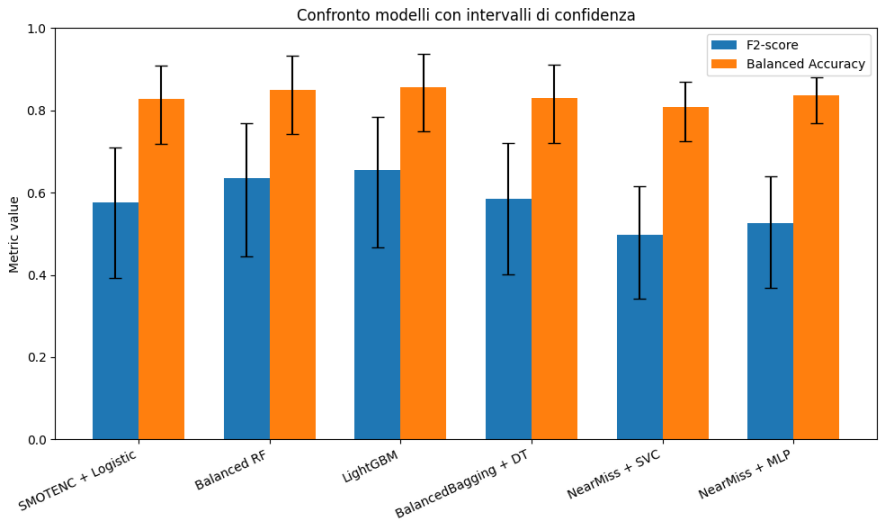

Figure 6.4: Comparison of model performance on the validation set. Bars represent F2-score and Balanced Accuracy with 95% confidence intervals.

Statistical validation via the Wilcoxon test on correct predictions confirmed that balanced ensemble methods significantly outperform simpler approaches, such as Logistic Regression combined with SMOTENC ($p_{\text{corr}} < 0.05$).

6.1.3 Functional Outcome: Barthel Index Prediction

The prediction of the post-operative Barthel Index (BI) further corroborated the superiority of tree-based and boosting algorithms. As illustrated in Figure 6.5, the most significant predictors include baseline functional scores and age.

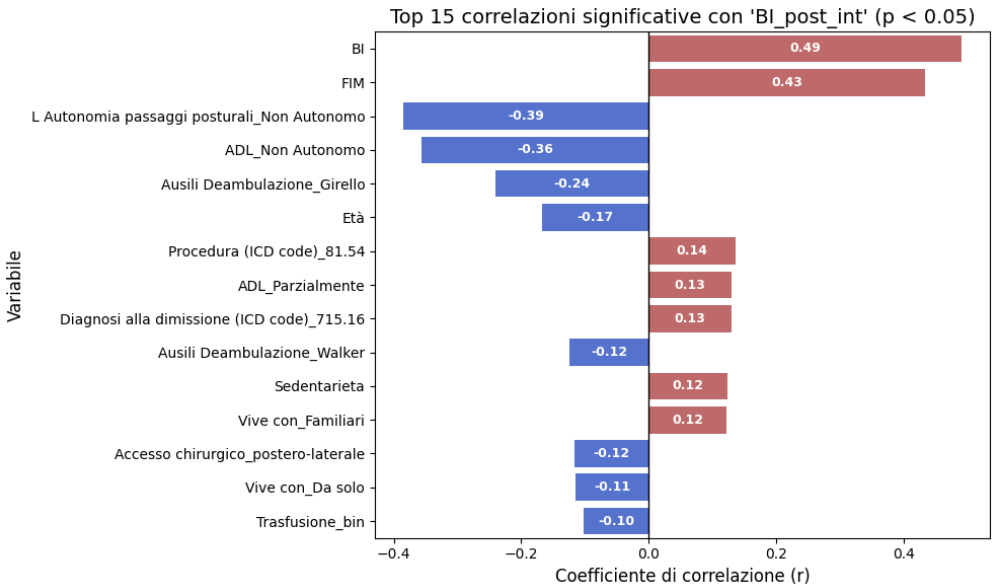

Figure 6.5: Top 15 significant correlations with *BI_post_int* ($p < 0.05$). Red bars indicate positive correlations, while blue bars indicate negative correlations.

LightGBM achieved the highest coefficient of determination ($R^2 = 0.340$) with an MAE of 2.50 points, followed closely by Random Forest ($R^2 = 0.328$). Detailed performance metrics for the validation set are presented in Table 6.3.

Table 6.3: Model performance on the validation set (with imputation)

Model	**MAE [95% CI]**	**RMSE [95% CI]**	R^2 **[95% CI]**	**MAE/SD**	**RMSE/SD**
Ridge Regression (α = 100)	2.69 [2.30, 3.20]	4.85 [3.23, 6.72]	0.295 [0.140, 0.431]	0.467	0.840
Lasso Regression (α = 0.01)	2.75 [2.36, 3.23]	4.88 [3.42, 6.68]	0.284 [0.086, 0.413]	0.477	0.846
Random Forest (optimized)	2.45 [2.05, 2.95]	4.73 [3.06, 6.67]	0.328 [0.132, 0.528]	0.424	0.820
LightGBM (optimized)	2.50 [2.12, 3.00]	4.69 [3.15, 6.70]	0.340 [0.114, 0.549]	0.434	0.812
Decision Tree (optimized)	2.56 [2.13, 3.07]	5.09 [3.34, 7.16]	0.222 [-0.090, 0.486]	0.443	0.882
SVR (optimized)	2.56 [2.17, 3.09]	4.79 [3.18, 6.72]	0.311 [0.117, 0.483]	0.443	0.830
MLP (optimized)	3.21 [2.72, 3.80]	5.87 [4.44, 7.61]	-0.035 [-0.674, 0.241]	0.557	1.017

In contrast, the Multi-Layer Perceptron (MLP) exhibited poor generalization, with a negative $R^2 = -0.035$ and the highest MAE (3.21). This result is consistent with the limited sample size relative to the complexity of deep learning architectures. The relative accuracy and correlation of these models are visually summarized in the Taylor Diagram in Figure 6.6.

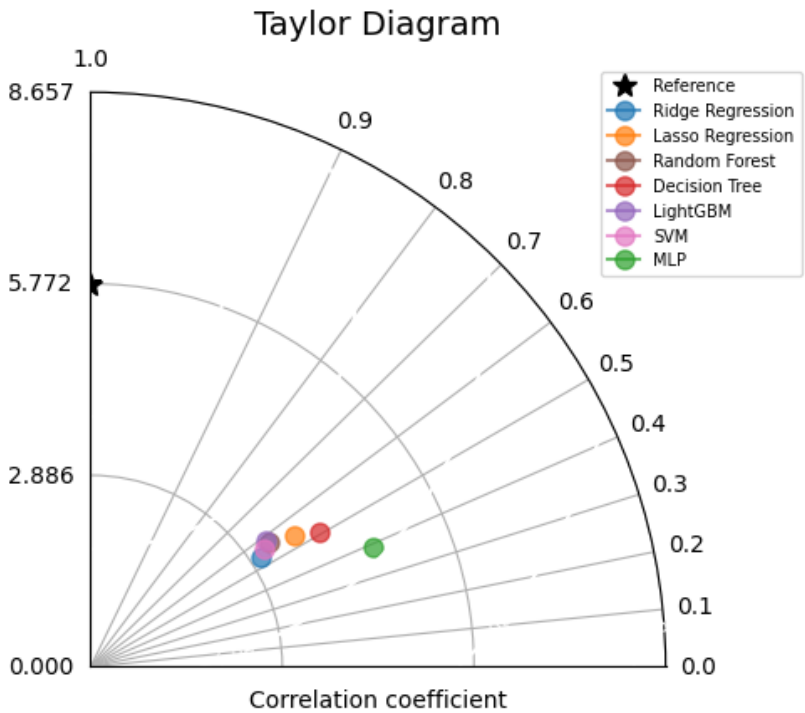

Figure 6.6: Taylor Diagram of predictive models for the Barthel Index. The reference point (black star) represents observed values.

Moreover, the ratios MAE/SD and RMSE/SD, approximately in the range 0.42–0.43 for the best-performing models, indicate that the prediction error accounts for less than half of the natural variability of the target. Consequently, model performance can be considered acceptable within the inherently multifactorial context of rehabilitation outcomes.

6.1.4 Comparative Visualization: Taylor Diagram

To provide a joint assessment of correlation, explained variance, and centered RMSE, a Taylor diagram was employed. This visualization highlights the relative performance of models in a unified framework.

The results show that LightGBM, SVR, and Random Forest are consistently positioned closer to the reference point, indicating superior agreement with observed data in terms of both accuracy and variance reproduction. Conversely, linear models tend to underestimate the dispersion of clinical outcomes, resulting in suboptimal alignment with the empirical distribution.

6.2 Statistical Validation Protocol

This section outlines the statistical validation protocol adopted to rigorously compare the performance of the proposed Machine Learning models. The primary objective is to assess whether the observed differences in evaluation metrics, including Mean Absolute Error (MAE) and F2-score, are statistically significant or attributable to random variation, thereby ensuring the methodological robustness of the study.

6.2.1 Methodology and Statistical Rationale

The selection of statistical tests was guided by both the distributional properties of the residuals and the experimental design, particularly with respect to the presence or absence of data imputation. Preliminary residual analysis revealed clear deviations from normality; consequently, parametric tests such as the Student's t-test were deliberately avoided. Instead, non-parametric alternatives were employed.

Specifically, the Wilcoxon signed-rank test was used for paired comparisons in the imputed data scenario (MICE), where all models were trained and evaluated on the same completed dataset, thus enabling point-wise comparison of predictions. Conversely, in the absence of imputation, the removal of records with missing values resulted in prediction sets of differing lengths across models. This precluded direct pairing and motivated the use of the Mann–Whitney U test for independent samples.

To control for the increased risk of Type I errors arising from multiple pairwise comparisons, the Benjamini–Hochberg procedure was systematically applied to control the False Discovery Rate (FDR). Only results with adjusted p-values ($p_{\text{corr}} < 0.05$) were considered statistically significant.

6.2.2 Comparison Results: Rehabilitation Duration

The statistical analysis consistently demonstrated the superiority of non-linear models over classical linear approaches. In the imputed scenario, the Wilcoxon test revealed highly significant differences ($p_{\text{corr}} < 0.001$) between Ridge Regression and non-linear models, including ensemble methods (Random Forest, LightGBM) and kernel-based approaches (SVR). Detailed results of these paired residual comparisons are presented in Table 6.4.

Table 6.4: Results of the paired Wilcoxon test on model residuals (with FDR correction)

Model 1	**Model 2**	**Statistic**	p**-value**	p_{corr} **(FDR)**	**Significant**
Ridge	Lasso	21219.0	0.023	0.045	True
Ridge	Random Forest	18691.0	< 0.001	< 0.001	True
Ridge	LightGBM	18389.0	< 0.001	< 0.001	True
Ridge	Decision Tree	20823.0	0.012	0.025	True
Ridge	SVR	18120.0	< 0.001	< 0.001	True
Ridge	MLP	19664.0	0.001	0.003	True
Lasso	Random Forest	18057.0	< 0.001	< 0.001	True
Lasso	LightGBM	17641.0	< 0.001	< 0.001	True
Lasso	Decision Tree	20339.0	0.005	0.012	True
Lasso	SVR	17457.0	< 0.001	< 0.001	True
Lasso	MLP	19239.0	< 0.001	0.001	True
Random Forest	LightGBM	24615.0	0.867	0.867	False
Random Forest	Decision Tree	23596.0	0.426	0.526	False
Random Forest	SVR	21622.0	0.044	0.076	False
Random Forest	MLP	22113.0	0.087	0.121	False
LightGBM	Decision Tree	24083.0	0.620	0.651	False
LightGBM	SVR	21903.0	0.065	0.098	False
LightGBM	MLP	21770.0	0.054	0.087	False
Decision Tree	SVR	23593.0	0.424	0.526	False
Decision Tree	MLP	23700.0	0.464	0.541	False
SVR	MLP	23864.0	0.528	0.584	False

No statistically significant differences were observed among the top-performing non-linear models (Random Forest, LightGBM, and SVR), with adjusted p-values exceeding 0.05. In the non-imputed scenario, the Mann–Whitney U test confirmed that differences between linear and tree-based models remained significant ($p_{\text{corr}} < 0.05$).

6.2.3 Comparison Results: Social Outcome Classification

For the classification task, the Wilcoxon signed-rank test was applied to the distributions of binary prediction outcomes (correct vs. incorrect) for each sample in the validation set. The results, summarized in Table 6.5, highlight the superior performance of balanced ensemble methods.

Table 6.5: Results of the paired Wilcoxon test for classification models on validation set samples (with FDR correction)

Model 1	**Model 2**	**Statistic**	p**-value**	p_{corr} **(FDR)**	**Significant**
LogReg_SMOTENC	BalancedRF	120.0	0.016	0.020	True
LogReg_SMOTENC	LightGBM	136.0	0.003	0.005	True
LogReg_SMOTENC	BalancedBagging_DT	472.5	0.763	0.763	False
LogReg_SMOTENC	SVC_NearMissv3	122.5	4.14×10^{-8}	1.24×10^{-7}	True
LogReg_SMOTENC	MLP_NearMissv3	243.0	1.53×10^{-6}	3.74×10^{-6}	True
BalancedRF	LightGBM	16.5	0.206	0.238	False
BalancedRF	BalancedBagging_DT	40.0	0.012	0.016	True
BalancedRF	SVC_NearMissv3	120.0	3.14×10^{-11}	2.36×10^{-10}	True
BalancedRF	MLP_NearMissv3	220.5	1.09×10^{-9}	4.08×10^{-9}	True
LightGBM	BalancedBagging_DT	33.0	0.001	0.002	True
LightGBM	SVC_NearMissv3	165.0	8.98×10^{-12}	1.35×10^{-10}	True
LightGBM	MLP_NearMissv3	234.5	1.55×10^{-10}	7.73×10^{-10}	True
BalancedBagging_DT	SVC_NearMissv3	532.5	1.74×10^{-6}	3.74×10^{-6}	True
BalancedBagging_DT	MLP_NearMissv3	612.0	1.13×10^{-5}	2.11×10^{-5}	True
SVC_NearMissv3	MLP_NearMissv3	288.0	0.612	0.656	False

In particular, LightGBM with `scale_pos_weight` demonstrated statistically significant improvements over Logistic Regression with SMOTENC ($p_{\text{corr}} = 0.005$). Furthermore, no significant difference was found between LightGBM and Balanced Random Forest ($p_{\text{corr}} = 0.238$), confirming both as robust solutions for handling class imbalance.

6.2.4 Comparison Results: Post-Intervention Barthel Index

The statistical evaluation of functional recovery prediction further emphasized the importance of model flexibility. Ensemble methods achieved significantly lower prediction errors compared to linear models ($p_{\text{corr}} < 0.05$), with specific pairwise results provided in Table 6.6.

Table 6.6: Wilcoxon paired signed-rank test results for model residuals (Barthel Index) with FDR correction

Model 1	**Model 2**	**Statistic**	p**-value**	p_{corr} **(FDR)**	**Significant**
Ridge	Lasso	24,018.0	0.592	0.592	False
Ridge	Random Forest	19,122.0	0.000368	0.00386	True
Ridge	LightGBM	20,909.0	0.01398	0.03670	True
Ridge	Decision Tree	20,640.0	0.00869	0.02607	True
Ridge	SVM (SVR)	21,663.0	0.04641	0.08860	False
Ridge	MLP	23,647.0	0.44411	0.54464	False
Lasso	Random Forest	19,587.0	0.00106	0.00740	True
Lasso	LightGBM	21,020.0	0.01689	0.03941	True
Lasso	Decision Tree	21,540.0	0.03867	0.08120	False
Lasso	SVM (SVR)	21,745.0	0.05226	0.09146	False
Lasso	MLP	23,448.0	0.37439	0.49139	False
Random Forest	LightGBM	23,010.0	0.24645	0.37156	False
Random Forest	Decision Tree	23,156.0	0.28517	0.39924	False
Random Forest	SVM (SVR)	23,015.0	0.24771	0.37156	False
Random Forest	MLP	18,897.0	0.000214	0.00386	True
LightGBM	Decision Tree	23,841.0	0.51870	0.54464	False
LightGBM	SVM (SVR)	23,821.0	0.51073	0.54464	False
LightGBM	MLP	19,720.0	0.00141	0.00740	True
Decision Tree	SVM (SVR)	23,824.0	0.51192	0.54464	False
Decision Tree	MLP	19,873.0	0.00195	0.00818	True
SVM (SVR)	MLP	20,541.0	0.00725	0.02537	True

Support Vector Regression (SVR) showed intermediate behavior, while Multi-Layer Per-

ceptron (MLP) exhibited significantly inferior generalization performance. Overall, the statistical evidence provides a consistent and rigorous basis for selecting LightGBM and Random Forest as the preferred models for clinical deployment.

6.3 Evaluation Framework and Performance Metrics

This section describes the comprehensive evaluation framework adopted to assess the effectiveness of the predictive models, ensuring both statistical validity and practical applicability of the results. The selection of evaluation tools was driven by the need to address class imbalance and non-normal data distributions, which are characteristic of real-world hospital datasets. The protocol is structured according to the nature of the task (regression or classification) and incorporates advanced analytical and visualization techniques to evaluate model robustness.

For regression tasks, including the prediction of rehabilitation duration and functional indices, model performance was quantified using multiple complementary metrics. The Mean Absolute Error (MAE) was used as a primary measure, providing an interpretable estimate of the average prediction error in the same unit as the target variable (e.g., days or Barthel points), with the best-performing models achieving an MAE of 2.20 days for length of stay. Furthermore, the Root Mean Squared Error (RMSE) was employed to penalize large deviations more heavily, while the coefficient of determination (R^2) was used to quantify the proportion of variance explained by the model, with observed values ranging between 0.25 and 0.27 for rehabilitation duration. To contextualize model error relative to the intrinsic variability of the data, normalized indices such as MAE/SD and RMSE/SD were computed. In particular, an RMSE/SD value greater than 1 indicates that the model error exceeds the natural variability of the target, suggesting poor generalization; this condition was observed in the external test set, where LightGBM yielded a value of 1.013. Additionally, 95% confidence intervals were estimated by non-parametric bootstrap with 1,000 replications, providing a robust quantification of uncertainty.

For classification tasks, such as social outcome and care burden prediction, the evaluation strategy accounted for severe class imbalance (e.g., 90.6% in the majority class for the social

outcome). Consequently, overall accuracy was deemed inadequate and potentially misleading. Instead, the F2-score was adopted as the primary optimization metric, as it assigns greater weight to recall relative to precision, thereby prioritizing the identification of positive cases. In alignment with this approach, Balanced Accuracy was used to assess the model's ability to correctly classify both classes by averaging sensitivity and specificity. Moreover, Receiver Operating Characteristic Area Under the Curve (ROC-AUC) and Average Precision (AP) were included to evaluate overall discriminative ability and the quality of probability ranking.

Model comparison was conducted within a rigorous statistical validation framework implemented using the `scikit-learn` library. Due to the non-normality of residuals, non-parametric tests were employed. Specifically, the Wilcoxon signed-rank test was applied to paired samples in the imputed (MICE) scenario, enabling point-wise comparison of absolute errors. Conversely, the Mann–Whitney U test was used for independent samples in the non-imputed scenario, where dataset sizes differed due to the exclusion of records with missing values. To control the false discovery rate associated with multiple comparisons, the Benjamini–Hochberg correction was systematically applied, with statistical significance defined as $p_{\text{corr}} < 0.05$.

The framework also incorporates advanced visualization techniques to support interpretation and assess clinical utility. The Taylor Diagram was used to jointly represent correlation, standard deviation, and centered RMSE within a single polar plot, enabling evaluation of the model's ability to reproduce the observed variability of the data. This analysis revealed a general tendency of models to underestimate the dispersion of outcomes. Calibration curves and the Brier Score were employed to assess the agreement between predicted probabilities and observed frequencies. In the external test set, a high Expected Calibration Error (ECE = 0.248) was observed, motivating the application of Platt scaling for recalibration. Furthermore, Decision Curve Analysis (DCA) was conducted to compute the Net Benefit, thereby evaluating the operational usefulness of the models relative to baseline strategies (i.e., treating all or no patients). The presence of negative Net Benefit values in the initial external evaluation indicated limited practical applicability without further adjustment.

Finally, robustness was assessed through explicit analysis of data distribution shifts. Adversarial validation was performed by training a Random Forest classifier to distinguish

between the training set (2019) and the external test set (2018); the resulting accuracy of 1.0 provided clear evidence of a critical data shift. This finding was further supported by Kolmogorov–Smirnov tests, which identified significant distributional differences in key variables (e.g., intervention duration and length of stay, with $p = 0.0$). In addition, an outlier analysis was conducted by identifying cases with absolute error greater than twice the MAE, allowing the isolation of instances characterized by extreme values (e.g., pre-intervention length of stay up to +6.6 standard deviations). Together, these analyses provide a comprehensive assessment of model reliability under varying data conditions.

Chapter 7

External Validation and Data Shift Analysis

7.1 Generalization, Data Shift, and External Validation

This section critically evaluates the ability of the developed models to generalize to unseen data, with particular emphasis on the presence of data shift and the associated performance degradation observed during external validation. Generalizability was assessed using an independent test set comprising 360 patients collected in 2018, representing a temporally distinct cohort relative to the training set (2019). To prevent data leakage, all preprocessing steps, including scaling, encoding, and MICE imputation, were estimated exclusively on the 2019 data and subsequently applied to the external test set.

The transition from internal to external validation revealed an immediate and substantial performance collapse, suggesting a structural discrepancy between the datasets. To formally investigate this hypothesis, a multi-level statistical protocol was implemented. First, a similarity test based on a permutation approach and distance measures between observations yielded a p-value of 0.0, indicating statistically significant distributional differences. Furthermore, adversarial validation was conducted by training a Random Forest classifier to distinguish whether a given record originated from the training set (2019) or the test set (2018). The model achieved an accuracy of 1.0, demonstrating perfect separability between the two cohorts in the feature space. In alignment with these findings, Kolmogorov–Smirnov tests applied to numerical variables identified the primary sources of distributional shift, with the most

pronounced differences observed in "intervention duration" and "length of stay" ($p < 0.05$), while other variables did not exhibit critical deviations.

The application of internally optimized models to the external cohort resulted in statistically and operationally inadequate performance across all evaluated outcomes. For the primary regression task (length of stay), the LightGBM model exhibited a marked decline, as detailed in Table 7.1. It achieved an MAE of 3.02 days (95% CI: 2.69–3.35), an R^2 of -0.027 (95% CI: -0.115, 0.061), and a generalization index (RMSE/SD) of 1.013. Notably, a negative R^2 and an RMSE/SD ratio greater than 1 indicate that the model error exceeds the natural variability of the target variable, implying performance inferior to a naive mean-based predictor.

Table 7.1: LightGBM model performance on the external test set

Metric	**Value**	**95% CI (Lower)**	**95% CI (Upper)**
MAE (days)	3.02	2.69	3.35
RMSE (days)	4.27	3.68	4.91
R^2	-0.027	-0.115	0.061

A similar deterioration was observed in the classification task (discharge destination). Although the LightGBM model retained moderate discriminative ability, as seen in the ROC and Precision-Recall curves (Figure 7.1), its practical utility was limited. Specifically, the F2-score decreased to 0.1866 (95% CI: 0.0463–0.3309), and sensitivity reached only 0.294. Additionally, the Expected Calibration Error (ECE) was 0.2480. Consequently, the combination of high calibration error and negative Net Benefit, as evidenced by the calibration and decision curves (Figure 7.2), indicates that the model would lead to suboptimal decision-making compared to baseline strategies due to systematic overestimation of risk.

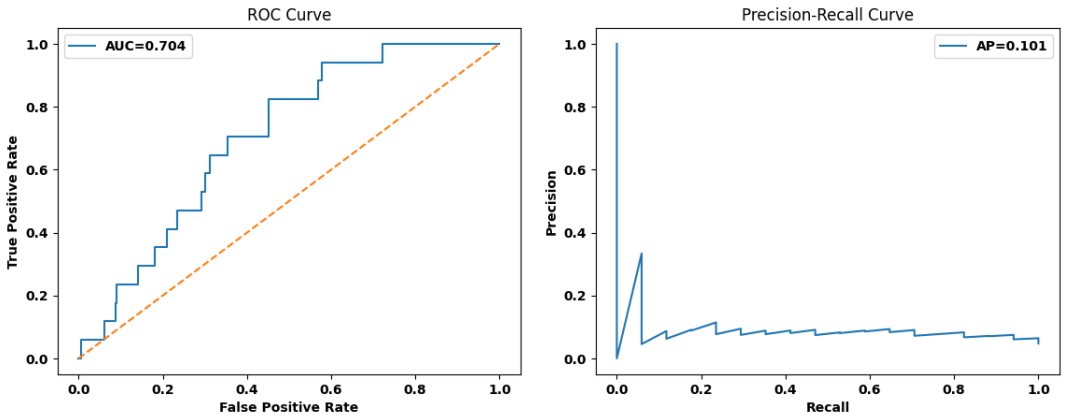

Figure 7.1: ROC and Precision-Recall curves on the external test set.

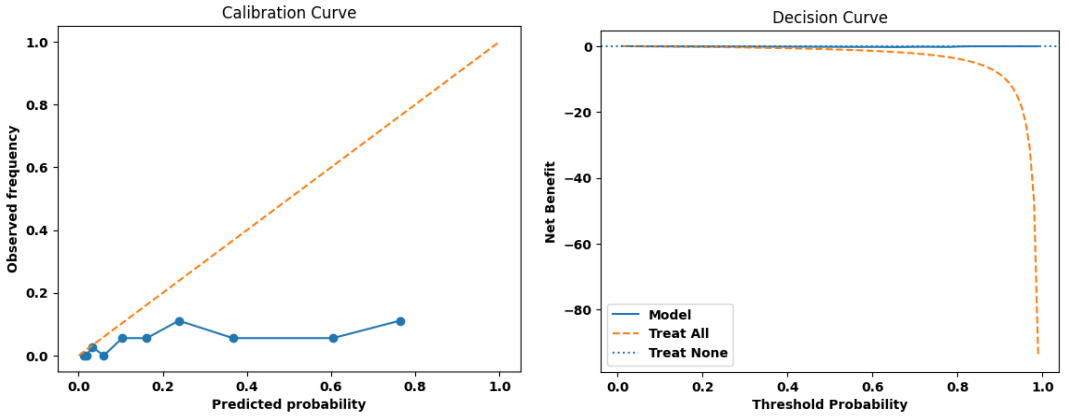

Figure 7.2: Calibration curve and Decision Curve Analysis.

The degradation extended to the prediction of post-intervention functional outcomes. The Random Forest model's performance on the external cohort is summarized in Table 7.2, yielding an MAE of 3.21 points (95% CI: 2.84–3.68) and an R^2 of -0.254, further confirming the loss of predictive reliability under distributional shift.

Table 7.2: Random Forest model performance on the external test set for post-intervention Barthel Index prediction

Metric	**Value**	**95% CI (Lower)**	**95% CI (Upper)**
MAE (BI points)	3.21	2.84	3.68
RMSE (BI points)	5.14	3.83	6.88
R^2	-0.254	-0.745	-0.093

Despite the statistical evidence of generalization failure, interpretability analyses based on SHAP values and feature importance indicated that the models consistently relied on stable variables, such as baseline functional status. Therefore, the observed performance collapse reflects the presence of both concept drift and label shift, as illustrated by the change in the mean of the target variable from 10.23 to 11.83 days.

7.2 Data Shift Detection and Characterization

This section examines the methodological and statistical procedures adopted to identify and quantify the data shift observed when transitioning from internal validation (2019 cohort) to external testing (2018 cohort, $n = 360$). The timely detection of this discrepancy was essential to justify the subsequent application of recalibration and model adaptation strategies.

First, a global similarity test was conducted using a permutation-based approach, yielding a p-value of $p = 0.0$, confirming that direct generalization was not feasible. Adversarial validation further quantified this degree of separability: a Random Forest classifier trained to distinguish between the 2018 and 2019 cohorts achieved an accuracy of 1.0 and an AUC ≈ 0.99, providing strong empirical evidence of a pronounced distributional shift.

To further isolate the variables contributing to this discrepancy, a univariate analysis was performed using the Kolmogorov–Smirnov (K-S) test on all numerical features. Statistically significant differences were primarily identified for “intervention duration” and “length of stay in the surgical ward” ($p < 0.05$).

The analysis enabled a more precise characterization of the observed data shift into three components: label shift (increase in mean rehabilitation duration), concept drift (evolution

of functional relationships), and calibration drift (Expected Calibration Error of 0.2480). In conclusion, these findings demonstrate that the learned relationships are not temporally stable, necessitating mitigation strategies such as Platt recalibration or cohort mixing to restore predictive performance.

Chapter 8

Data Shift Mitigation Techniques

8.1 Post-hoc Mitigation Strategies under Data Shift

This section analyzes the post-hoc strategies implemented to mitigate the substantial performance degradation observed during external validation (2018 cohort), where predictive models exhibited a marked collapse (e.g., $R^2 = -0.027$ for length of stay) due to data shift. These techniques aim to adjust the decision rule or recalibrate predicted probabilities while preserving the original model architecture.

For the classification task (social outcome), decision threshold optimization was conducted to address the limitations of the default threshold (0.5) in the presence of severe class imbalance (90.6% majority class) and distributional shift. A systematic search over the threshold interval was performed using two optimization criteria: maximization of the F2-score and enforcement of a minimum sensitivity constraint (≥ 0.85) combined with maximal Balanced Accuracy. Under the F2-score criterion, the optimal threshold was identified at $t = 0.22$, resulting in an increase in F2-score from 0.216 to 0.299 and in Balanced Accuracy from 0.582 to 0.677; however, this improvement was accompanied by an increase in false positives from 65 to 121. When prioritizing sensitivity ($t = 0.08$), sensitivity reached 0.94 and false negatives decreased from 11 to 1, at the cost of 200 false positives. Despite these improvements in classification metrics, the Net Benefit remained negative across all thresholds, indicating that threshold adjustment alone is insufficient to compensate for structural data shift.

To address miscalibration in predicted probabilities, recalibration techniques were applied using a 20% subset of the external test set. The original model exhibited poor calibration,

with an Expected Calibration Error (ECE) of 0.256 and a Brier score of 0.1365. Platt scaling, a parametric approach based on logistic regression applied to predicted probabilities, yielded a substantial improvement in reliability, reducing the ECE to 0.055 and the Brier score to 0.0618. When combined with an optimized threshold ($t \approx 0.04$), this approach increased the F2-score to 0.279 and the Balanced Accuracy to 0.656, while reducing false negatives from 9 to 3. In contrast, isotonic regression, a non-parametric alternative, achieved a similar ECE (0.055) but resulted in inferior classification performance (F2 $\approx$ 0.20), likely due to overfitting associated with the limited calibration sample size.

Furthermore, importance weighting was explored to mitigate covariate shift by reweighting training observations to approximate the external data distribution. A discriminative classifier (adversarial validation via logistic regression) was trained to distinguish between the internal set (IS) and the external set (OS), achieving an AUC of 0.99, thereby confirming substantial misalignment. Importance weights were computed as $w(x) = \frac{p_{OS}(x)}{p_{IS}(x)}$, with capping at the 95th percentile and normalization to ensure numerical stability.

The effectiveness of importance weighting varied across tasks. For rehabilitation duration, no improvement was observed: the MAE remained unchanged at 3.03, while R^2 deteriorated from -0.027 to -0.046. This failure is attributable to a pronounced label shift, as reflected by the change in the target mean (11.83 days in OS vs 10.23 in IS). For the social outcome, calibration improved (ECE reduced from 0.256 to 0.161) and Net Benefit increased (from -0.0210 to -0.0006 at threshold 0.1), albeit with a decrease in F2-score from 0.216 to 0.177. For the Barthel Index prediction, performance further deteriorated, with R^2 declining from -0.252 to -0.420, exacerbated by a strong ceiling effect in the target variable.

Overall, these analyses demonstrate that post-hoc techniques, including threshold optimization, probability recalibration, and importance weighting, are effective in addressing aspects of covariate shift and improving probabilistic reliability. However, they fail to resolve label shift and concept drift, which involve changes in the target distribution and in the relationship between covariates and outcomes. In such scenarios, the only effective strategy identified in this study is cohort mixing (i.e., training on a combined dataset), which restored predictive performance, yielding a positive R^2 value (0.253) for length of stay.

8.2 Distributional Correction Strategies under Data Shift

This section examines advanced distributional correction methodologies implemented to address the pronounced data shift identified during external validation of the predictive system. The presence of a statistically significant discrepancy between the training cohort (2019) and the external test cohort (2018) was confirmed through a global similarity test ($p = 0.0$) and adversarial validation achieving an accuracy of 1.0. These findings necessitated the exploration of techniques aimed at realigning the underlying data distributions.

The first strategy involved importance weighting to mitigate covariate shift by reweighting observations from the internal training set (IS) to better approximate the distribution of the external set (OS). The implementation relied on a discriminative pipeline based on logistic regression with feature standardization, trained to distinguish between IS and OS samples. The predicted probability $P(y = 1 \mid x)$ was used to estimate the density ratio and assign weights according to $w(x) = \frac{p_{OS}(x)}{p_{IS}(x)}$. To ensure numerical stability and prevent extreme observations from dominating the loss function, weights were normalized such that their sum matched the original training set size, and a capping procedure at the 95th percentile was applied.

The effectiveness of importance weighting varied across tasks. For rehabilitation duration, no improvement was observed, with R^2 remaining negative and slightly deteriorating (from -0.027 to -0.046). This outcome was attributed to a pronounced label shift, as the mean of the target variable in the external set (11.83 days) differed substantially from that of the training set (10.23 days), with extreme values not represented in the original data. For the social outcome, the method led to improved calibration, with the Expected Calibration Error (ECE) decreasing from 0.256 to 0.161, and an increase in operational utility, as the Net Benefit approached neutrality at threshold 0.1 (from -0.0210 to -0.0006). Overall, these results indicate that importance weighting is effective in correcting probability misalignment due to covariate shift, but insufficient in addressing structural changes such as label shift and concept drift.

Given the limitations of post-hoc correction techniques, a cohort mixing strategy was subsequently implemented. This approach directly modifies the learning space by exposing

the model to the variability of both cohorts, thereby enabling the extraction of more robust and generalizable patterns. The experimental protocol involved concatenating the datasets into a unified population of 1,963 observations, which was then randomly split into training (80%) and test (20%) subsets. The models were retrained using the same optimized architectures previously identified, including LightGBM and Random Forest.

The resulting performance improvements were substantial across all tasks. For rehabilitation duration, R^2 increased from negative values to 0.253, accompanied by a reduction in MAE from 2.99 to 2.25 days. In the classification task (social outcome), all evaluation metrics improved significantly, with ROC-AUC reaching 0.84 (compared to 0.68), F2-score increasing to 0.487, and Balanced Accuracy to 0.754. Notably, the model achieved a positive Net Benefit (0.0071 at threshold 0.1) for the first time. Similarly, for the Barthel Index prediction, R^2 became positive (0.237), overcoming the limitations previously associated with the observed ceiling effect.

In conclusion, the comparative analysis demonstrates that, in the presence of complex data shift involving both covariates and target distributions, recalibration and reweighting techniques are inherently limited. In contrast, cohort mixing emerges as the only effective strategy in this study, highlighting that the integration of multi-year data provides a more robust solution than post-hoc model adaptation for ensuring transferability and practical applicability of the predictive system.

8.3 Model Adaptation and Mitigation Strategies under Data Shift

This section examines the model adaptation methodologies and mitigation strategies implemented to address the data shift identified during external validation. Despite strong internal performance, the transition to the 2018 cohort resulted in a substantial degradation of predictive accuracy (e.g., negative R^2 for length of stay), necessitating both post-hoc and structural interventions to restore the operational utility of the system.

For the classification task (social outcome), decision threshold optimization was performed to overcome the limitations of the default threshold (0.5) under conditions of class

imbalance and distributional shift. A systematic search over the interval $[0.01, 0.99]$ was conducted using two optimization criteria: maximization of the F2-score and enforcement of a minimum sensitivity constraint (≥ 0.85) combined with maximal Balanced Accuracy. Under the F2-score criterion, the optimal threshold was identified at $t = 0.22$, yielding an increase in F2-score from 0.216 to 0.299 and in Balanced Accuracy from 0.582 to 0.677. When prioritizing sensitivity, the optimal threshold decreased to $t = 0.08$, achieving a sensitivity of 0.94 and reducing false negatives from 11 to 1, albeit with 200 false positives. This optimization strategy reflects the prioritization of minimizing false negatives; however, the Net Benefit remained negative across all thresholds, indicating that threshold adjustment alone is insufficient to address structural data shift.

Probability recalibration was subsequently applied to correct systematic misestimation in predicted probabilities. The original model exhibited poor calibration on the external test set, with an Expected Calibration Error (ECE) of 0.256 and a Brier score of 0.1365. Two recalibration techniques were evaluated on a validation subset corresponding to 20% of the external test set. Platt scaling, a parametric approach based on logistic regression applied to predicted probabilities, substantially improved calibration, reducing the ECE to 0.055 and the Brier score to 0.0618. When combined with an optimized threshold ($t \approx 0.04$), this approach increased the F2-score to 0.279. In contrast, isotonic regression, a non-parametric method, achieved a comparable ECE (0.055) but resulted in inferior classification performance (F2 ≈ 0.20), likely due to overfitting given the limited size of the calibration subset.

To mitigate covariate shift, importance weighting was explored by reweighting observations from the internal training set (IS) to approximate the external distribution (OS). A discriminative classifier (logistic regression with standardization) was trained to distinguish IS from OS, achieving an AUC of 0.99 and confirming substantial distributional misalignment. Importance weights were computed as $w(x) = \frac{p_{OS}(x)}{p_{IS}(x)}$, with capping at the 95th percentile and normalization applied to ensure numerical stability. The effectiveness of this approach varied across tasks. For rehabilitation duration, no improvement was observed, with R^2 deteriorating from -0.027 to -0.046, primarily due to a pronounced label shift (mean target value of 11.83 days in OS versus 10.23 in IS). For the social outcome, calibration improved (ECE reduced from 0.256 to 0.161) and Net Benefit increased (from -0.0210 to -0.0006 at threshold 0.1).

Conversely, for the Barthel Index prediction, performance worsened, with R^2 decreasing from -0.252 to -0.420.

Given the limitations of post-hoc techniques in addressing structural changes in the data-generating process, a cohort mixing strategy was implemented to directly modify the learning space. The datasets were concatenated into a unified sample (n = 1,963), randomly split into training (80%) and test (20%) subsets, and the models were retrained using the previously optimized architectures. This approach yielded substantial improvements across all tasks. For rehabilitation duration, R^2 increased from negative values to 0.253. For the social outcome, all performance metrics improved significantly, with ROC-AUC reaching 0.84 and F2-score increasing to 0.487; notably, the Net Benefit became positive (0.0071 at threshold 0.1). Similarly, for the Barthel Index prediction, R^2 became positive (0.237).

Finally, a feature selection strategy based on robustness was explored to reduce potential overfitting to patterns specific to the 2019 cohort. The methodology involved selecting the top 10 features based on importance estimated during internal validation and retraining the model accordingly. However, results remained largely unchanged for the regression task (length of stay), with R^2 still negative (-0.007). For the social outcome, a moderate improvement in Balanced Accuracy was observed (from 0.582 to 0.632), although without a corresponding increase in operational utility.

In conclusion, the analyses demonstrate that, in the presence of complex data shift involving both label shift and concept drift, recalibration and reweighting techniques are insufficient to restore model performance. In alignment with these findings, cohort mixing emerges as the only effective strategy, indicating that integrating multi-year data is more effective than post-hoc adaptation for ensuring model generalizability and practical applicability.

Chapter 9

Explainability and Sustainability

9.1 Explainability and Model Interpretability

This section describes the methodological framework adopted to ensure transparency and interpretability of the predictive models, a critical requirement for clinical validation and integration into hospital decision-making workflows within a Clinical Decision Support System (CDSS). The objective of Explainable Artificial Intelligence (XAI) in this study is twofold: to identify the key determinants associated with rehabilitation outcomes and to assess the consistency of the learned patterns with established domain knowledge.

The interpretability strategy was tailored according to model complexity. Linear models were leveraged for their intrinsic interpretability, while for ensemble methods such as Random Forest and LightGBM, global interpretability was obtained through feature importance measures (MDI and Gain). For the primary regression task (rehabilitation length of stay), a strong convergence across models was observed. As shown in Figure 9.1, LightGBM identifies intervention duration and functional indices (FIM and BI) as the most influential variables.

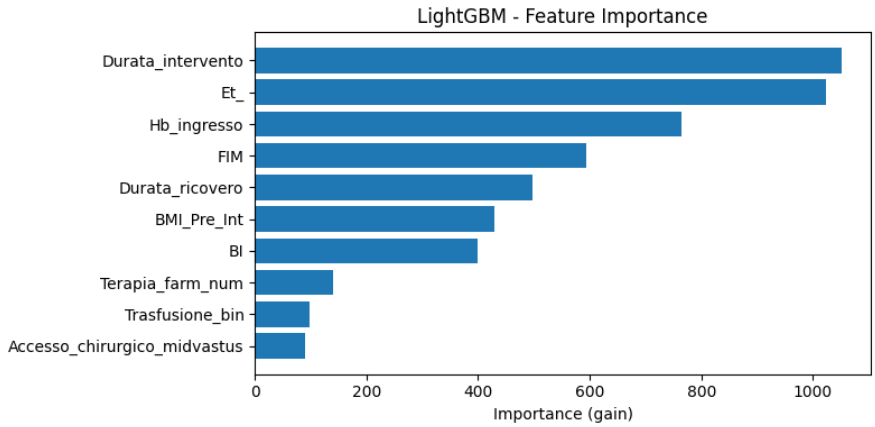

Figure 9.1: **Variable importance in the LightGBM model.** Relative feature importance calculated based on Gain.

SHAP (SHapley Additive exPlanations) analysis further confirmed these patterns, providing a rigorous quantification of the marginal contribution of each feature. Figure 9.2 illustrates how longer intervention times and advanced age (positive SHAP values) increase the predicted length of stay.

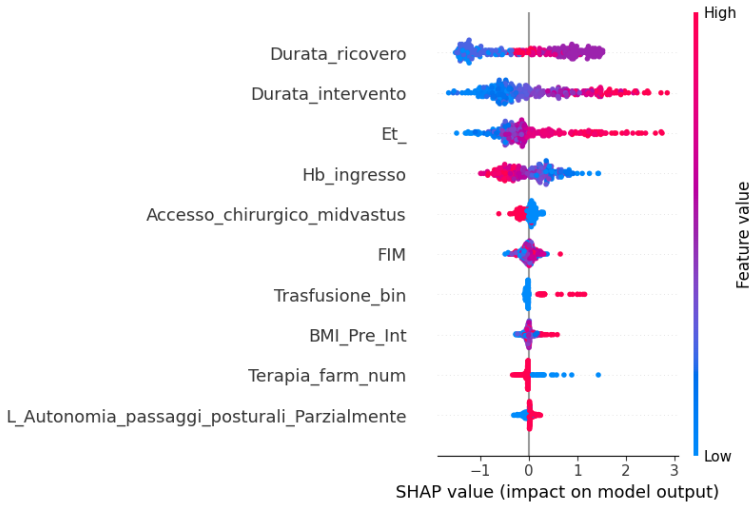

Figure 9.2: **SHAP value distribution for the LightGBM model.** Distribution of mean variable importance according to SHAP values.

For the classification task (discharge destination), the XAI analysis highlighted the dominant role of socio-environmental variables. Grouped permutation importance in the Random

Forest model (Figure 9.3) shows that the "living arrangement" group accounts for more than one-third of the total model importance.

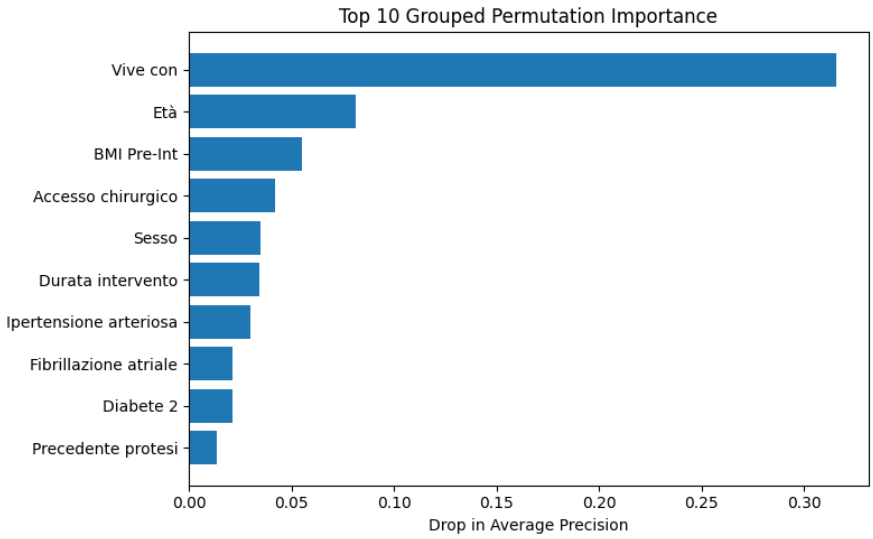

Figure 9.3: **Top 10 Grouped Permutation Importance.** The ten most relevant feature groups for social outcome classification.

SHAP analysis for the social outcome (Figure 9.4) revealed that the condition "living alone" produced strongly positive contributions, acting as a primary driver for predictions of reduced residential autonomy.

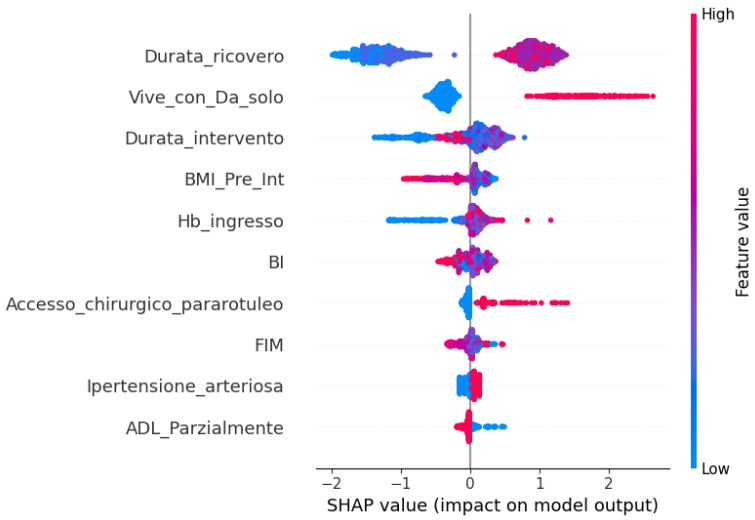

Figure 9.4: **SHAP value distribution for the social outcome.** Impact of variables on the LightGBM model for discharge destination.

Regarding the post-intervention Barthel Index, models consistently identified baseline functional status as the dominant predictor (Figure 9.5). SHAP analysis (Figure 9.6) further illustrates that low baseline BI values substantially reduced predicted outcomes, while advanced age acted as a negative modifier.

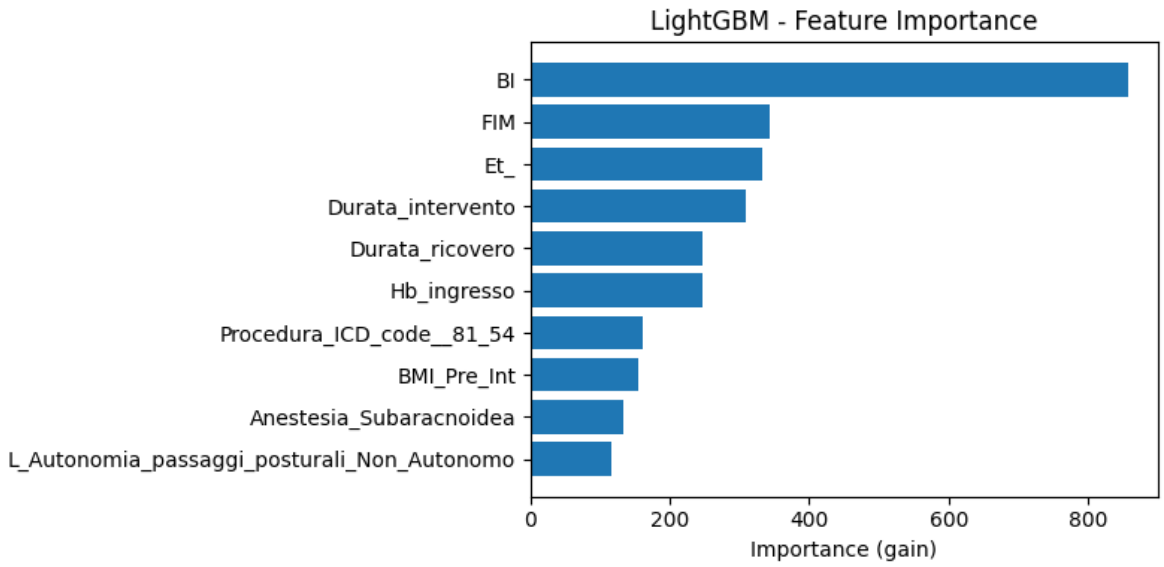

Figure 9.5: **Variable importance for the post-intervention Barthel Index.** Relative feature importance in the LightGBM model based on Gain.

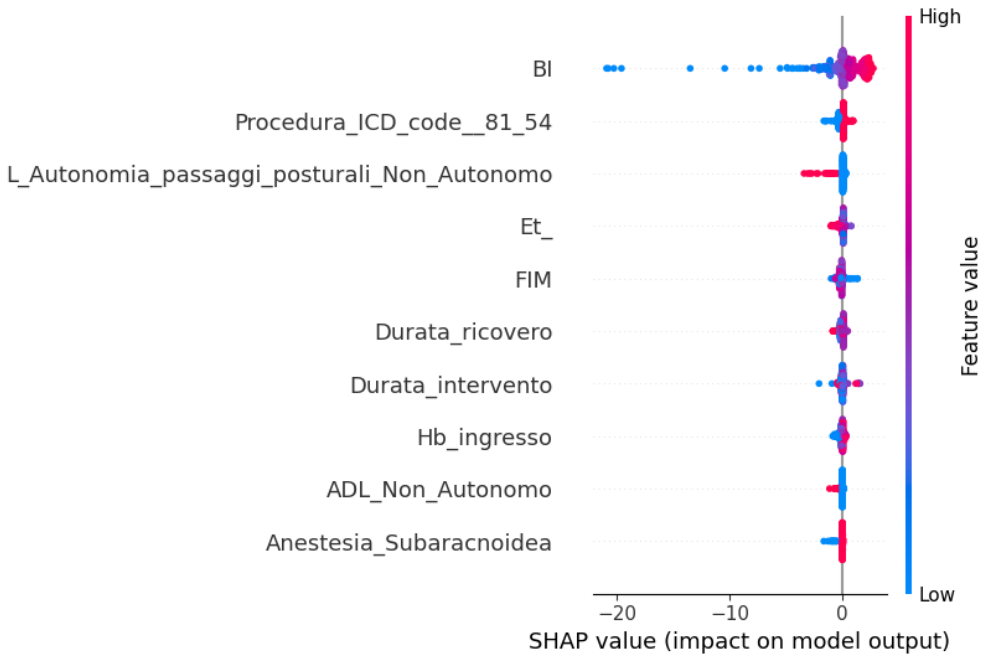

Figure 9.6: **SHAP analysis for the post-intervention Barthel Index.** Feature impact on the LightGBM model. Color indicates feature value (Red: High, Blue: Low); the X-axis represents the model prediction contribution.

9.2 Computational Efficiency and Environmental Sustainability

This section evaluates the computational and energy impact of the proposed framework in accordance with Green AI principles. All predictive models were designed to ensure a limited environmental impact. Computations were performed on an Intel i7 CPU (8 cores), with overall emissions for the full training cycle estimated below 0.005 kg CO_2 equivalent.

Inference efficiency was high across all architectures; the prediction time for a single patient was below 0.05 seconds. The prioritization of tree-based models (Random Forest and LightGBM) over deep learning (MLP) was supported by their superior performance-to-efficiency ratio. For instance, the MLP required up to 289 minutes for training without yielding proportional predictive benefits.

9.3 Integration into Clinical Workflow and Adoption

The system is designed to operate during the immediate post-operative phase, utilizing baseline clinical data and surgical parameters already present in the Electronic Health Records (EHR).

Clinicians, surgeons, and hospital administrators can leverage multidimensional outputs, length of stay, social autonomy, and functional recovery, to optimize resource allocation and care pathway planning. For example, identification of patients with low admission Barthel Index scores (which SHAP analysis shows as a primary negative driver of recovery) allows for early risk stratification.

To ensure long-term clinical robustness, the system includes a data shift management protocol. This involves systematic detection of dataset drift via Adversarial Validation and Kolmogorov–Smirnov tests, followed by cohort mixing strategies to restore predictive utility. Currently at Technology Readiness Level (TRL) 4, the system requires semi-annual monitoring and annual explainability validation to maintain its maturity and reliability in real-world healthcare settings.

Chapter 10

Conclusions and Future Work

10.1 Model Validation, Data Shift, and Performance Assessment

This section synthesizes the numerical and methodological evidence emerging from internal validation, external testing, and distributional shift mitigation. The study analyzed a cohort of 1,710 patients (year 2019), expanding the feature space to 77 predictive variables in order to model three primary outcomes using both Machine Learning and Deep Learning architectures.

10.1.1 Internal Validation and the Impact of Preprocessing

Internal validation, conducted using stratified 5-fold cross-validation, demonstrated the superiority of non-linear models over linear baselines. For the primary outcome of length of stay, Support Vector Regression (SVR) achieved an $R^2 = 0.265$ with a Mean Absolute Error (MAE) of 2.26 days, while LightGBM obtained an $R^2 = 0.250$ and MAE = 2.20 days, representing the most accurate models. Furthermore, the application of Multiple Imputation by Chained Equations (MICE) by the `IterativeImputer` (scikit-learn) proved critical, yielding an approximate increase of +0.05 in R^2 for SVR compared to the complete-case approach.

For the social outcome classification task, LightGBM with `scale_pos_weight` achieved the best performance, with an F_2-score of 0.656 and a Balanced Accuracy of 0.856. The selection of the F_2-score as the primary evaluation metric was driven by the requirement to prioritize recall over precision. Regarding post-operative Barthel Index prediction, ensemble

models outperformed alternatives, with LightGBM reaching an $R^2 = 0.340$ and MAE of 2.50 points. Conversely, the Multi-Layer Perceptron (MLP) exhibited poor generalization, with a negative R^2 (-0.035).

10.1.2 Statistical Comparison of Model Performance

To assess the statistical significance of performance differences, non-parametric tests were employed. Specifically, the Wilcoxon signed-rank test was applied to paired samples in the MICE scenario, while the Mann–Whitney U test was used for independent samples without imputation. The results confirmed statistically significant differences ($p_{\text{corr}} < 0.05$ after Benjamini–Hochberg False Discovery Rate correction) between ensemble models (Random Forest, LightGBM) and linear models (Ridge, Lasso). In contrast, no statistically significant difference was observed between LightGBM and Random Forest ($p_{\text{corr}} = 0.867$), indicating stable performance among top-performing models.

10.1.3 External Validation and Data Shift Detection

Evaluation on an independent cohort (n = 360, year 2018) revealed a substantial performance collapse. A permutation-based statistical similarity test yielded $p = 0.0$, indicating a significant distributional discrepancy. Moreover, a Random Forest classifier trained for Adversarial Validation—tasked with distinguishing between 2018 and 2019 samples—achieved an Accuracy of 1.0, confirming that the datasets are perfectly separable. Univariate analysis using the Kolmogorov–Smirnov test identified distributional shifts in the variables *procedure duration* and *length of stay* ($p < 0.05$). Consequently, predictive performance deteriorated markedly: for rehabilitation length of stay, R^2 decreased to -0.027, with an RMSE/SD ratio of 1.013, indicating performance inferior to a naive mean predictor.

10.1.4 Mitigation Strategies and Final Results

Several post-hoc strategies were evaluated to restore clinical utility. Threshold optimization (t = 0.22) improved the F_2-score to 0.299, while Platt Scaling reduced the Expected Calibration Error (ECE) from 0.256 to 0.055. However, the Net Benefit remained negative, limiting

practical applicability. In contrast, the most effective approach was cohort mixing, whereby models were trained on a combined dataset (n = 1,963), effectively addressing both concept drift and label shift. This strategy restored predictive performance, with R^2 for length of stay returning to a positive value (0.253) and the ROC-AUC for the social outcome increasing to 0.84.

10.1.5 Interpretability and Sustainability

Model interpretability analysis by SHAP values and feature importance (Gain) confirmed the internal consistency of the learned relationships. The most influential predictors were admission Barthel Index, procedure duration, and age. Finally, the system demonstrated high computational sustainability: the entire training cycle generated emissions below 0.005 kg CO_2 equivalent, while inference time per patient remained under 0.05 seconds, supporting real-time deployment within clinical workflows.

10.2 Limitations and Future Directions

Despite providing a robust methodological framework for the application of Machine Learning to orthopedic rehabilitation prognosis, this study presents several intrinsic limitations that constrain its immediate clinical applicability and delineate directions for future research. The following analysis integrates quantitative results and statistical evidence derived from the validation procedures.

10.2.1 Study Limitations

A primary limitation concerns the vulnerability of the models to temporal data shift and the resulting lack of generalizability. External validation on an independent cohort (year 2018) revealed a substantial degradation in performance. Specifically, the coefficient of determination (R^2) for length of stay decreased from 0.250 (internal validation, LightGBM) to -0.027 in external testing, while for the post-operative Barthel Index the R^2 dropped to -0.254. In alignment with these findings, multiple statistical tests confirmed the presence of structural

distributional differences between cohorts. A permutation-based similarity test yielded $p = 0.0$, indicating extreme divergence. Furthermore, Adversarial Validation, implemented by a Random Forest classifier trained to distinguish between the two cohorts, achieved an Accuracy of 1.0, demonstrating perfect separability in the feature space. Univariate analysis by the Kolmogorov–Smirnov test identified significant shifts in the variables *procedure duration* and *length of stay* ($p < 0.05$), suggesting temporal changes in underlying clinical processes.

A second limitation pertains to the intrinsic quality of the gold standard labels. The labeling process lacks transparency, as no information is available regarding inter-rater agreement or the data collection protocol (manual versus automated extraction from Electronic Health Records). Consequently, the absence of explicit validation procedures, such as consensus mechanisms or majority voting, introduces potential noise in the target variables, thereby limiting the upper bound of achievable model performance.

Additionally, the explanatory power of the current feature set remains limited. Despite the inclusion of 77 predictive variables, model performance did not exceed an R^2 of 0.30 across most tasks. This suggests that key latent factors influencing length of stay and functional recovery are not captured in the dataset. Such factors may include organizational aspects of hospital workflows, unobserved patient-level characteristics, or procedural details not encoded within ICD-based representations.

Finally, Deep Learning architectures proved inadequate in this setting. The Multi-Layer Perceptron (MLP), despite optimization using RandomizedSearchCV, underperformed compared to tree-based ensemble methods. For the post-operative Barthel Index prediction, the MLP yielded a negative R^2 (-0.035) and the highest MAE (3.21), indicating poor generalization. This result is consistent with the limited sample size (approximately 1,700 patients), where the higher complexity of neural networks may induce overfitting or instability relative to more robust ensemble approaches like LightGBM and Random Forest.

10.2.2 Future Directions and Improvement Strategies

In light of the observed limitations, several directions for future development are proposed. First, transitioning toward multi-center and multi-year learning frameworks is essential. Post-hoc correction techniques, including Platt Scaling and Importance Weighting, were insufficient

to address concept drift. Conversely, the cohort mixing strategy—training on a combined dataset (n = 1,963), successfully restored clinical utility, improving the R^2 for length of stay to 0.253 and increasing the ROC-AUC for the social outcome to 0.84. Consequently, future work should focus on integrating data from multiple orthopedic centers to develop models with broader generalizability.

Furthermore, the implementation of a structured maintenance protocol is required to mitigate temporal degradation in predictive performance. This includes semi-annual model retraining using the most recent data to reduce drift effects. In parallel, annual explainability audits should be conducted to verify the stability and consistency of SHAP values. For example, it is necessary to ensure that higher hemoglobin levels at admission continue to exhibit negative SHAP contributions to length-of-stay predictions, thereby preventing the model from capturing transient statistical artifacts.

Finally, expanding the feature space and incorporating advanced domain adaptation techniques represent critical steps for performance improvement. The inclusion of more granular socio-demographic variables and hospital workflow indicators (e.g., bed occupancy rates at admission) may help surpass the current $R^2 = 0.30$ threshold. Additionally, more sophisticated domain adaptation methods should be explored to enable model transferability across different surgical units with minimal recalibration, particularly in light of the elevated Expected Calibration Error (ECE = 0.248) observed during external testing.

In conclusion, although the system is currently positioned at Technology Readiness Level (TRL) 4, addressing the data shift problem through the integration of heterogeneous cohorts constitutes a necessary step toward achieving TRL 6 and enabling routine deployment in orthopedic clinical practice.

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